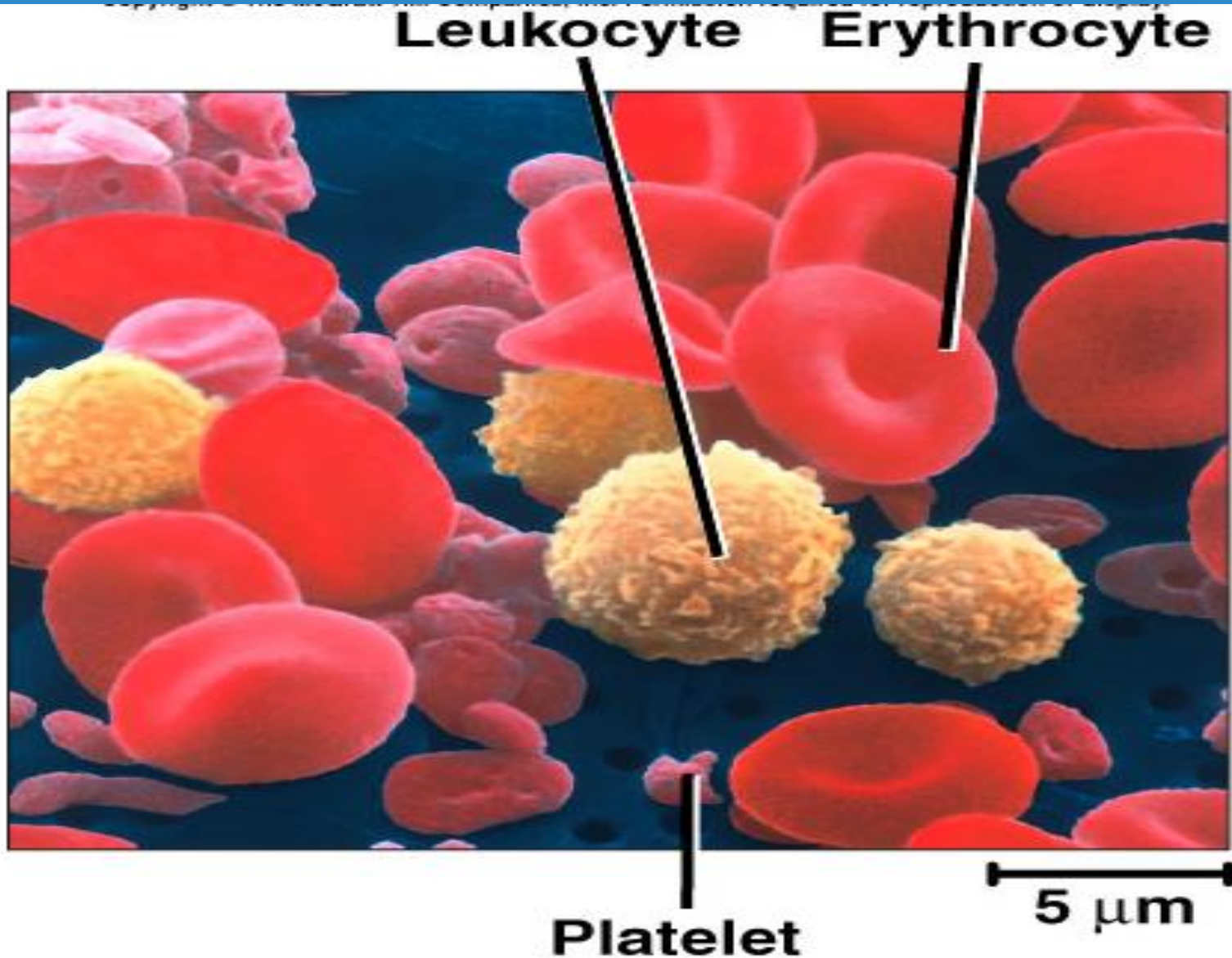


The Blood is the Life



Fluids of the Body

- Cells of the body are serviced by 2 fluids
 - blood
 - composed of plasma and a variety of cells
 - transports nutrients and wastes
 - interstitial fluid
 - bathes the cells of the body
- Nutrients and oxygen diffuse from the blood into the interstitial fluid & then into the cells
- Wastes move in the reverse direction
- Hematology is study of blood and blood disorders

Functions of Blood

- Transportation
 - O₂, CO₂, metabolic wastes, nutrients, heat & hormones
- Regulation
 - helps regulate pH through buffers
 - helps regulate body temperature
 - coolant properties of water
 - vasodilatation of surface vessels dump heat
 - helps regulate water content of cells by interactions with dissolved ions and proteins
- Protection from disease & loss of blood

Properties of Blood

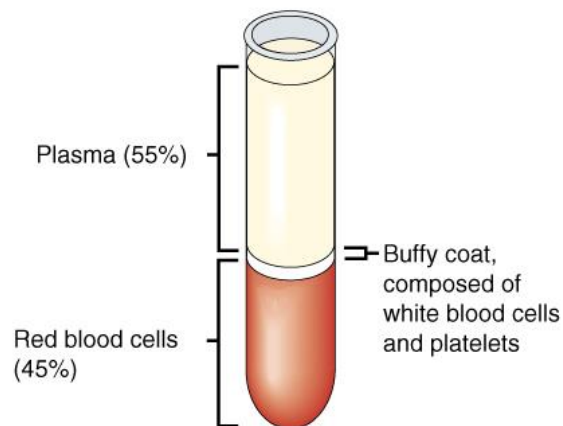
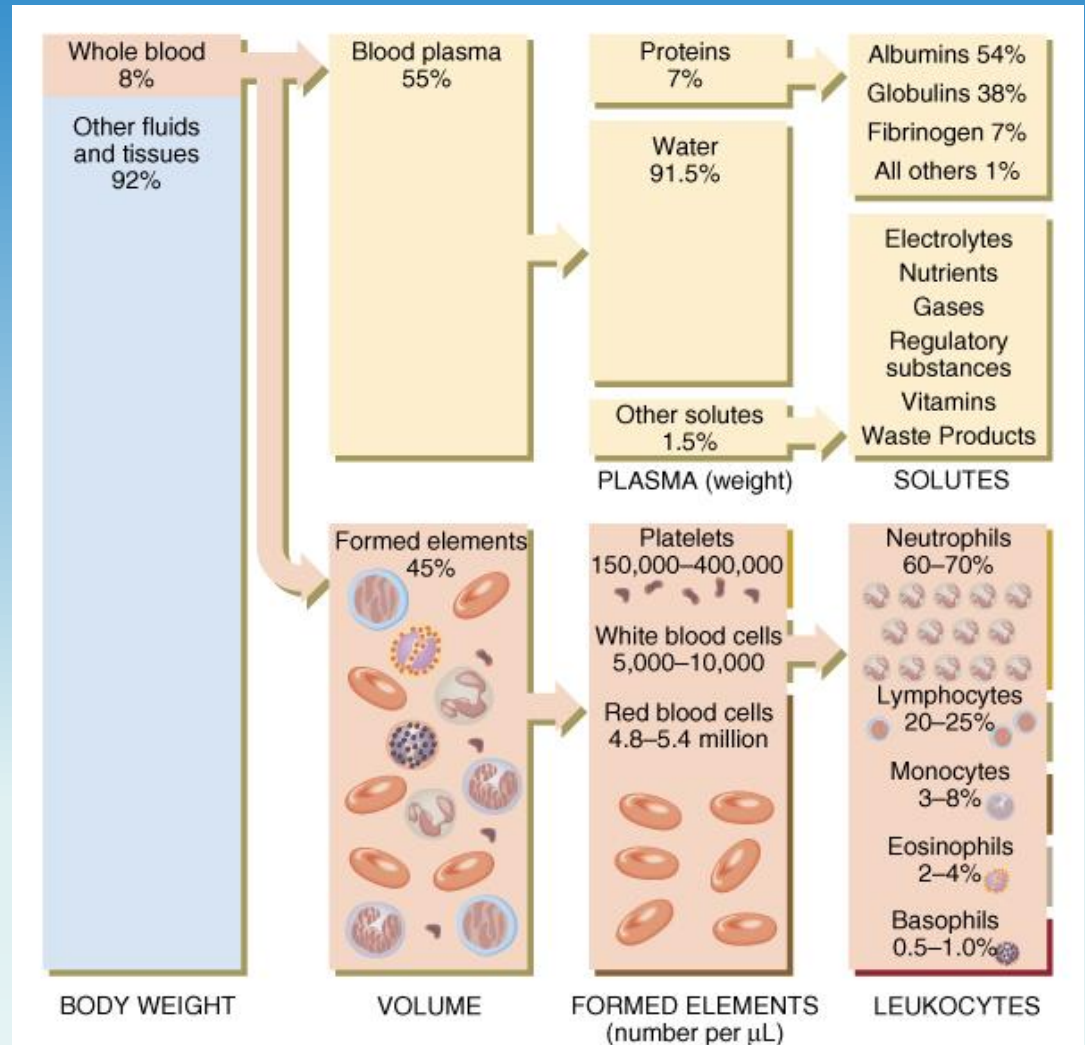
- Adults have 4-6 L of blood
 - plasma, a clear extracellular fluid
 - formed elements (blood cells and platelets)
- Properties of viscosity (resistance to flow)
- Thicker (more viscous) than water and flows more slowly than water
- Temperature of 100.4 degrees F
- pH 7.4 (7.35-7.45)
- 8 % of total body weight
- Blood volume
 - 5 to 6 liters in average male
 - 4 to 5 liters in average female
 - hormonal negative feedback systems maintain constant blood volume and osmotic pressure
 - molarity of dissolved particles)
 - if too high, fluid absorption into the blood causes high BP
 - if too low, fluid remains in the tissues causing edema
 - one cause is deficiency of plasma protein due to diet or disease

Techniques of Blood Sampling

- Venipuncture
 - sample taken from vein with hypodermic needle & syringe
 - median cubital vein
 - why not stick an artery?
 - less pressure
 - closer to the surface
- Finger or heel stick
 - common technique for diabetics to monitor daily blood sugar
 - method used for infants

Components of Blood

- Hematocrit
 - 55% plasma
 - 45% cells
 - 99% RBCs
 - < 1% WBCs and platelets



Plasma and Plasma Proteins

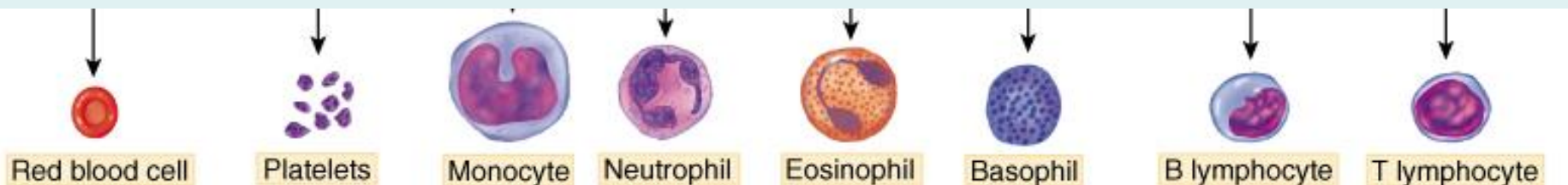
- Plasma is a mixture of water (>90%), proteins (7%), enzymes, nutrients, wastes, hormones, electrolytes, and gases (2% total)
 - if allowed to clot, what remains is called serum
- 3 major categories of plasma proteins
 - albumins are most abundant plasma protein
 - contributes to viscosity and osmolarity and influences blood pressure, flow and fluid balance
 - globulins (antibodies) provide immune system defenses
 - alpha, beta and gamma globulins
 - fibrinogen is precursor of fibrin threads that help form blood clots
- All plasma proteins formed by liver except globulins (produced by plasma cells descended from B lymphocytes)

Nonprotein Components of Plasma

- Plasma contains nitrogenous compounds
 - amino acids from dietary protein or tissue breakdown
 - nitrogenous wastes(urea) are toxic end products of catabolism
 - normally removed from the blood by the kidneys
- Nutrients (glucose, vitamins, fats, minerals, etc)
- Some O₂ and CO₂ are transported in plasma
- Many electrolytes are found in plasma
 - sodium makes up 90% of plasma cations accounting for more of blood's osmolarity than any other solute

Formed Elements of Blood

- Red blood cells (erythrocytes)
- White blood cells (leukocytes)
 - granular leukocytes
 - neutrophils
 - eosinophils
 - basophils
 - agranular leukocytes
 - lymphocytes = T cells, B cells, and natural killer cells
 - monocytes
- Platelets (special cell fragments)



Hematocrit

- Percentage of blood occupied by cells
 - female normal range
 - 38 - 46% (average of 42%)
 - male normal range
 - 40 - 54% (average of 46%)
 - testosterone
- Anemia
 - not enough RBCs or not enough hemoglobin
- Polycythemia
 - too many RBCs (over 65%)
 - dehydration, tissue hypoxia, blood doping in athletes

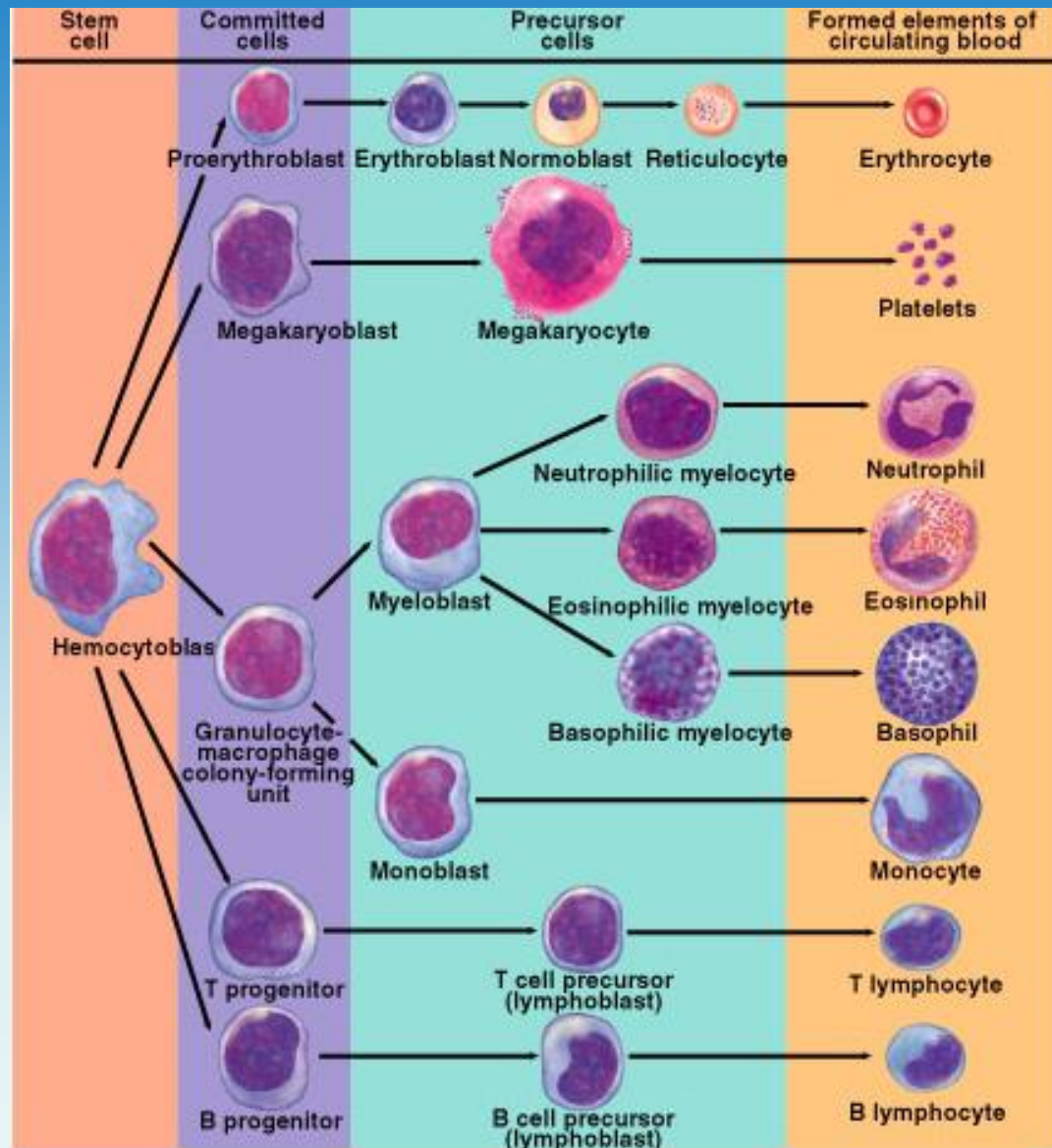
Blood Doping

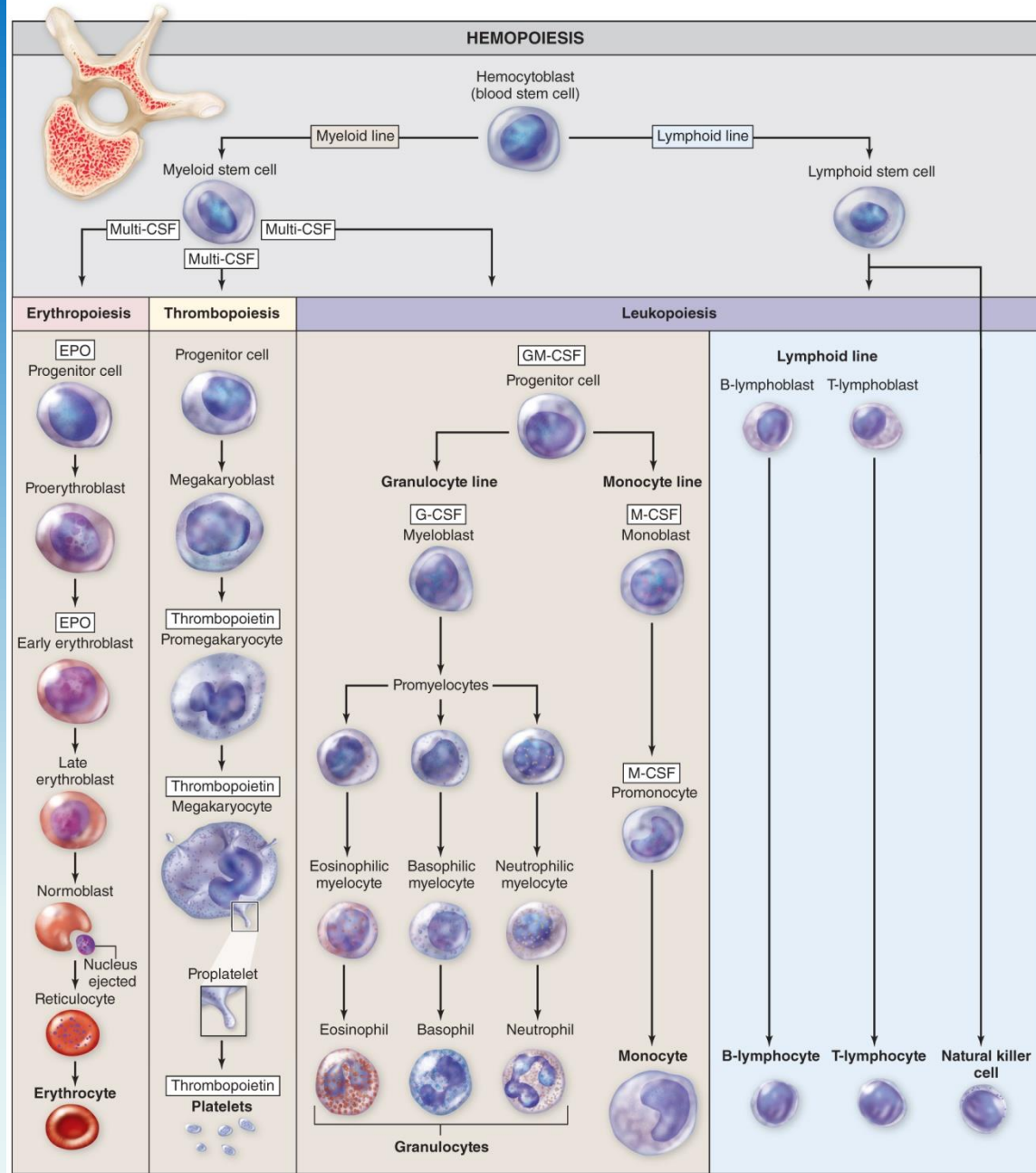
- Injecting previously stored RBC's before an athletic event
 - more cells available to deliver oxygen to tissues
- Dangerous
 - increases blood viscosity
 - forces heart to work harder
- Banned by Olympic committee

Blood Cell Production (Hemopoiesis)

- Most cells need continual replacement due to life spans of hours to weeks
- Hemopoietic tissues produce blood cells
 - yolk sac in vertebrate embryo produce stem cells that colonize fetal bone marrow, liver, spleen & thymus
 - liver stops producing blood cells at birth, but spleen and thymus remain involved with WBC production
 - lymphoid hemopoiesis occurs in widely distributed lymphoid tissues (thymus, tonsils, lymph nodes, spleen & peyers patches in intestines)
 - red bone marrow produces RBCs, WBCs and platelets
 - stem cells called hemocytoblasts multiply continually & are pluripotent (capable of differentiating into multiple cell lines)
 - committed cells are destined to continue down one specific cell line
 - In adults only in flatbones like sternum, ribs, skull, pelvis, head of femur and humerus
- Stimulated by erythropoietin, thrombopoietin & colony-stimulating factors (CSFs)

Hemopoiesis





Stages of Blood Cell Formation

- Pluripotent stem cells
 - .1% of red marrow cells
 - replenish themselves as they differentiate into either myeloid or lymphoid stem cells
- Myeloid stem cell line of development continues:
 - progenitor cells(colony-forming units) no longer can divide and are specialized to form specific cell types
 - example: CFU-E develops eventually into only red blood cells
 - next generation is blast cells
 - have recognizable histological characteristics
 - develop within several divisions into mature cell types
- Lymphoid stem cell line of development
 - pre-B cells & prothymocytes finish their develop into B & T lymphocytes in the lymphatic tissue after leaving the red marrow

Hemopoietic Growth Factors

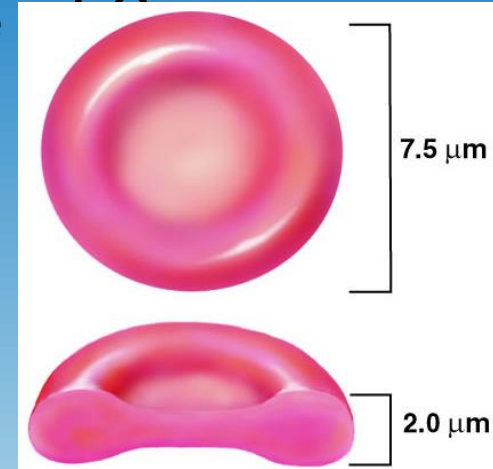
- Regulate differentiation & proliferation
- Erythropoietin (EPO)
 - produced by the kidneys increase RBC precursors
- Thrombopoietin (TPO)
 - hormone from liver stimulates platelet formation
- Cytokines are local hormones of bone marrow
 - produced by some marrow cells to stimulate proliferation in other marrow cells
 - colony-stimulating factor (CSF) & interleukin stimulate WBC production

Medical Uses of Growth Factors

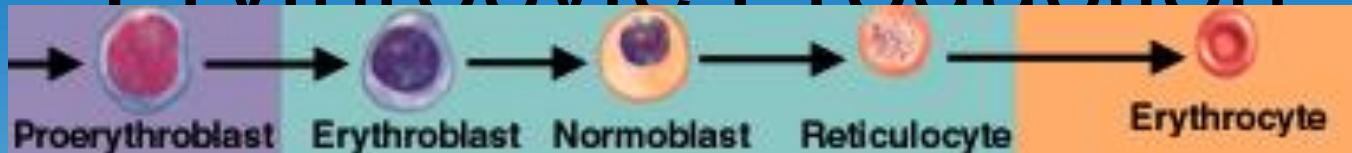
- Available through recombinant DNA technology
 - recombinant erythropoietin (EPO) very effective in treating decreased RBC production of end-stage kidney disease
 - other products given to stimulate WBC formation in cancer patients receiving chemotherapy which kills bone marrow
 - granulocyte-macrophage colony-stimulating factor
 - granulocyte colony stimulating factor
 - thrombopoietin helps prevent platelet depletion during chemotherapy

Erythrocytes (RBCs)

- New RBC's enter circulation at the rate of 2 million/second
- Disc-shaped cell with thick rim
 - 7.5 μm diameter & 2.0 μm thick at rim
- Major function is gas transport
 - lost all organelles during maturation so has increased surface area/volume ratio
 - increases diffusion rate of substances in & out of cell
 - 33% of cytoplasm is hemoglobin (Hb)
 - O₂ delivery to tissue and CO₂ transport back to lungs
 - contains enzyme, carbonic anhydrase (CAH)
 - produces carbonic acid from CO₂ and water
 - important role in gas transport & pH balance



Erythrocyte Production



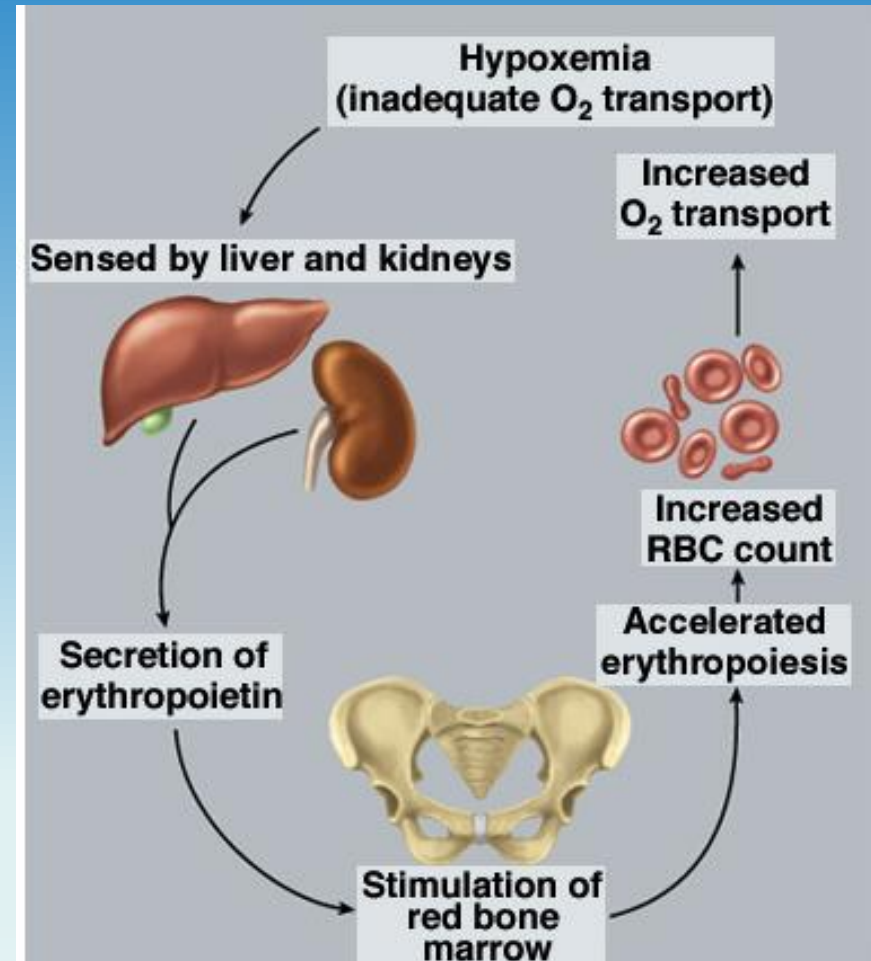
- Erythropoiesis produces 2.5 million RBCs/second from stem cells (hemocytoblasts) in bone marrow
- First committed cell is proerythroblast
 - has receptors for erythropoietin (EPO) from kidneys
- Erythroblasts multiply & synthesize hemoglobin
- Normoblasts discard their nucleus to form a reticulocyte
 - named for fine network of endoplasmic reticulum
 - enters bloodstream as 0.5 to 1.5% of circulating RBCs
- Development takes 3-5 days & involves
 - reduction in cell size, increase in cell number, synthesis of hemoglobin & loss of nucleus
 - blood loss speeds up the process increasing reticulocyte count

Normal Reticulocyte Count

- Should be .5 to 1.5% of the circulating RBC's
- Low count in an anemic person might indicate bone marrow problem
 - leukemia, nutritional deficiency or failure of red bone marrow to respond to erythropoietin stimulation
- High count might indicate recent blood loss or successful iron therapy

Erythrocyte Homeostasis

- Classic negative feedback control
 - drop in RBC count causes hypoxemia to kidneys
 - EPO production $\uparrow$
 - stimulation of bone marrow
 - RBC count $\uparrow$ in 3-4 days
- Stimulus for erythropoiesis
 - low levels of atmospheric O_2
 - increase in exercise
 - hemorrhaging



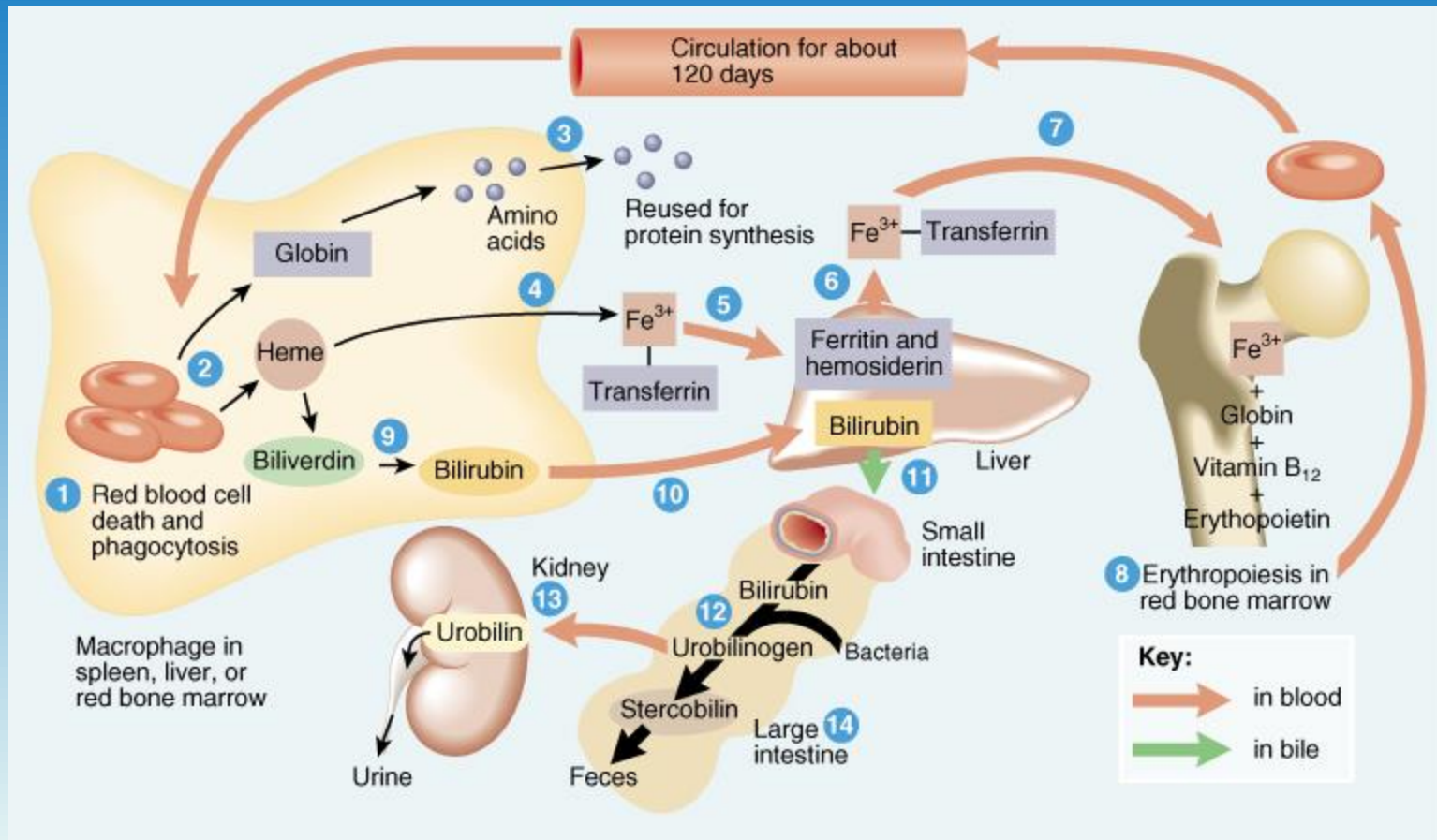
RBC Life Cycle

- RBCs live only 120 days
 - wear out from bending to fit through capillaries
 - no repair possible due to lack of organelles
- Worn out cells removed by fixed macrophages in spleen & liver
- Breakdown products are recycled

Nutritional Needs for Erythropoiesis

- Iron is key nutritional requirement for erythropoiesis
 - lost daily through urine, feces, and bleeding
 - men 0.9 mg/day and women 1.7 mg/day
 - low absorption rate requires consumption of 5-20 mg/day
 - dietary iron in 2 forms: ferric (Fe^{+3}) & ferrous (Fe^{+2})
 - stomach acid converts Fe^{+3} to absorbable Fe^{+2}
 - gastroferritin from stomach binds Fe^{+2} & transports it to intestine
 - absorbed into blood & binds to transferrin to travel
 - » bone marrow uses to make hemoglobin, muscle used to make myoglobin and all cells use to make cytochromes in mitochondria
 - liver binds surplus to apoferritin to create ferritin for storage
- B12 & folic acid (for rapid cell division) and C & copper for cofactors for enzymes synthesizing RBCs

Recycling of Hemoglobin

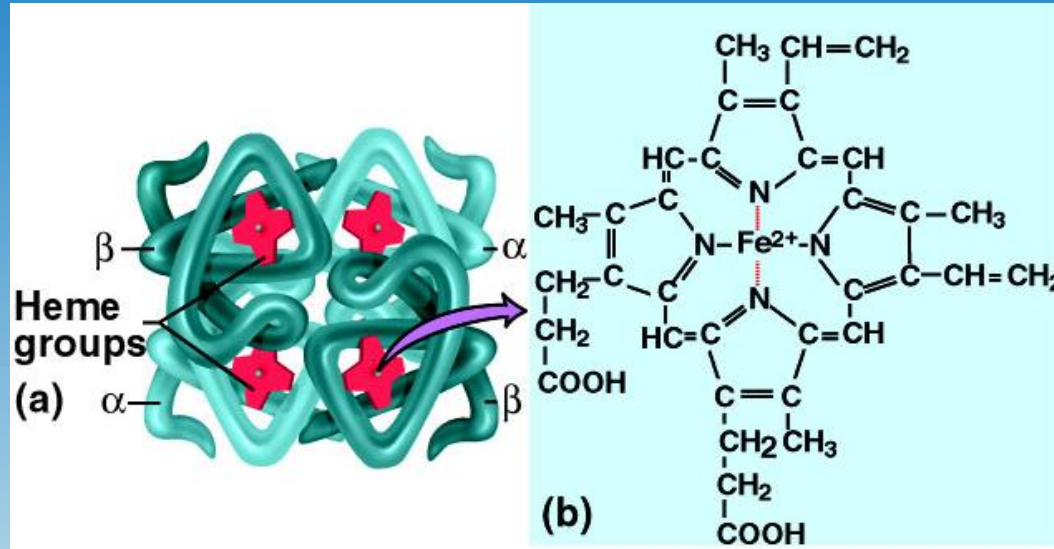


- In macrophages of liver or spleen
 - globin portion broken down into amino acids & recycled
 - heme portion split into iron (Fe^{+3}) and biliverdin (green pigment)

Fate of Components of Heme

- Iron(Fe^{+3})
 - transported in blood attached to transferrin protein
 - stored in liver, muscle or spleen
 - attached to ferritin or hemosiderin protein
 - in bone marrow being used for hemoglobin synthesis
- Biliverdin (green) converted to bilirubin (yellow)
 - bilirubin secreted by liver into bile
 - converted to urobilinogen then stercobilin (brown pigment in feces) by bacteria of large intestine
 - if reabsorbed from intestines into blood is converted to a yellow pigment, urobilin and excreted in urine

Hemoglobin Structure

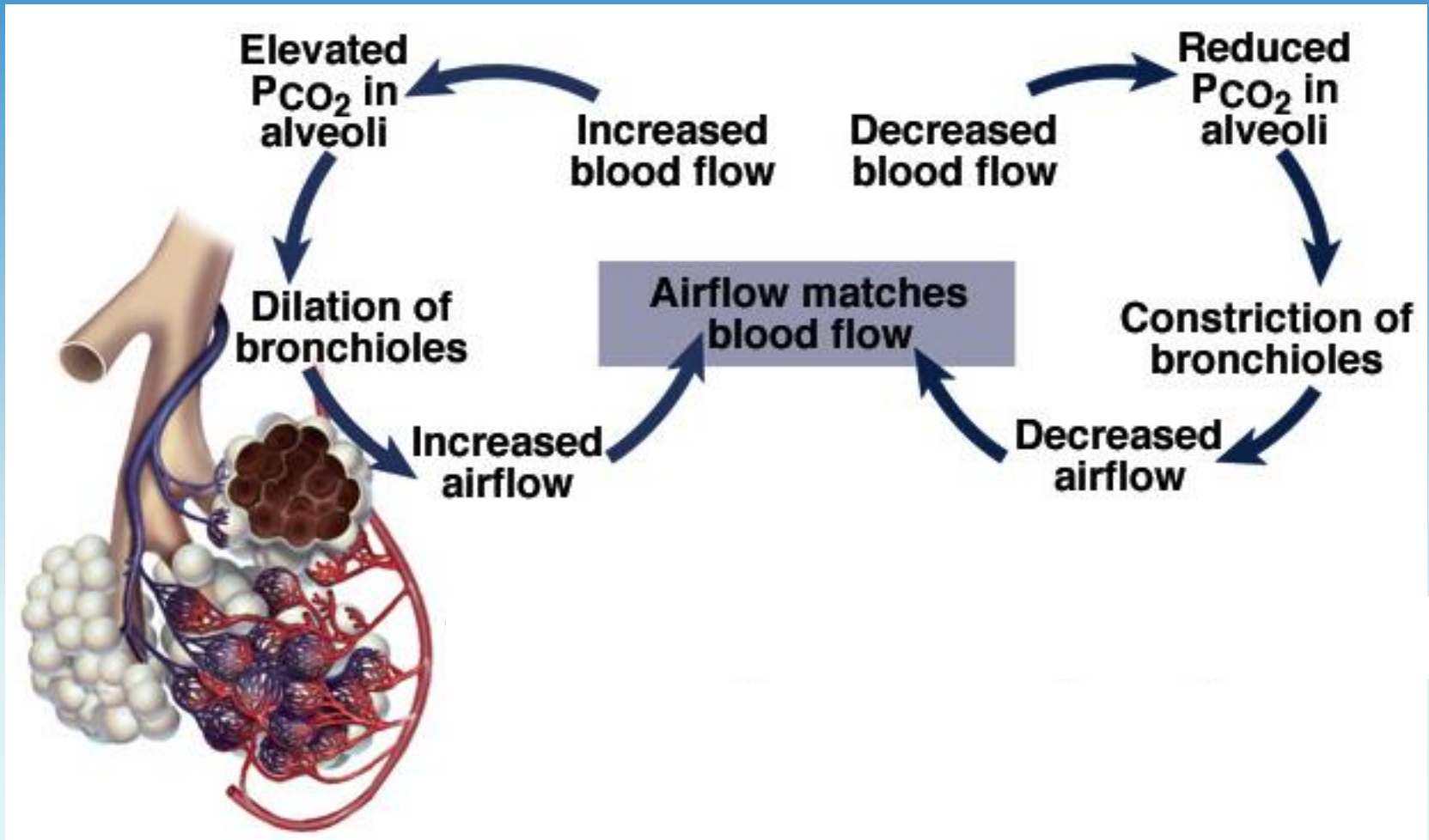


- Hemoglobin consists of 4 protein chains called globins (2 alpha & 2 beta)
- Each protein chain is conjugated with a heme group which binds oxygen to ferrous ion (Fe^{+2})
- Hemoglobin molecule can carry four O_2
- Fetal hemoglobin has gamma instead of beta chains

Transport of O₂, CO₂ and Nitric Oxide

- Each hemoglobin molecule can carry 4 oxygen molecules from lungs to tissue cells
- Hemoglobin transports 23% of total CO₂ waste from tissue cells to lungs for release
 - combines with amino acids in globin portion of Hb
- Hemoglobin transports nitric oxide & super nitric oxide helping to regulate BP
 - hemoglobin picks up nitric oxide (NO) & super nitric oxide (SNO) & transport it to & from the lungs
 - NO causing vasodilation is released in the lungs due to CO₂ release
 - NO causing vasodilation is released in the tissues due to O₂ release

Ventilation Adjusts to Changes in Perfusion



Erythrocytes and Hemoglobin

- RBC count & hemoglobin concentration indicate the amount of oxygen the blood can carry
 - hematocrit(packed cell volume) is % of blood composed of cells
 - men 42-52% cells; women 37-48% cells
 - hemoglobin concentration of whole blood
 - men 13-18g/dL; women 12-16g/dL
 - RBC count
 - men 4.6-6.2 million/ μ L; women 4.2-5.4 million/ μ L
- Values are lower in women
 - androgens stimulate RBC production
 - women have periodic menstrual losses

Erythrocyte Disorders

- Polycythemia is an excess of RBC
 - primary polycythemia is due to cancer of erythropoietic cell line in the red bone marrow
 - RBC count as high as 11 million/ μ L; hematocrit of 80%
 - secondary polycythemia from dehydration, emphysema, high altitude, or physical conditioning
 - RBC count only up to 8 million/ μ L
- Dangers of polycythemia
 - increased blood volume, pressure and viscosity can lead to embolism, stroke or heart failure

Anemia - Deficiency of RBCs or

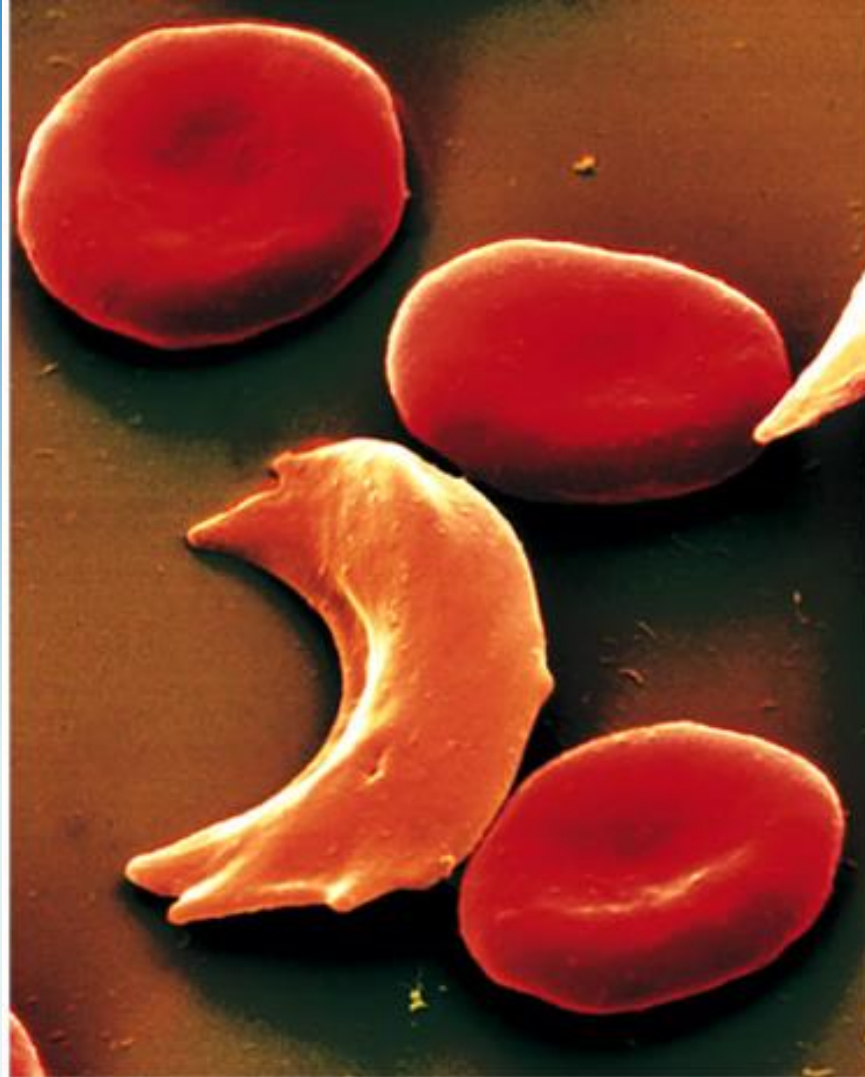
Hb

- Causes of anemia
 - inadequate erythropoiesis or hemoglobin synthesis
 - inadequate vitamin B12 from poor nutrition or lack of intrinsic factor from glands of the stomach (pernicious anemia)
 - iron-deficiency anemia
 - kidney failure & insufficient erythropoietin hormone
 - aplastic anemia is complete cessation (cause unknown)
 - hemorrhagic anemias from loss of blood
 - hemolytic anemias from RBC destruction
 - thalassemia = hereditary deficiency of hemoglobin
- Effects of anemia
 - tissue hypoxia and necrosis (short of breath & lethargic)
 - low blood osmolarity (tissue edema)
 - low blood viscosity (heart races & pressure drops)

Sickle-cell Anemia (SCA)

- Genetic defect in hemoglobin molecule (Hb-S) that changes 2 amino acids
 - at low very O₂ levels, RBC is deformed by changes in hemoglobin molecule within the RBC
 - sickle-shaped cells rupture easily = causing anemia & clots
- Found among populations in malaria belt
 - Mediterranean Europe, sub-Saharan Africa & Asia
- HbS gene persists despite harmful effects
 - increased resistance to malaria because RBC membranes leak K⁺ & lowered levels of K⁺ kill the parasite infecting the red blood cells (heterozygote advantage)

Sickle-Cell Diseased Erythrocyte



WBC Anatomy and Types

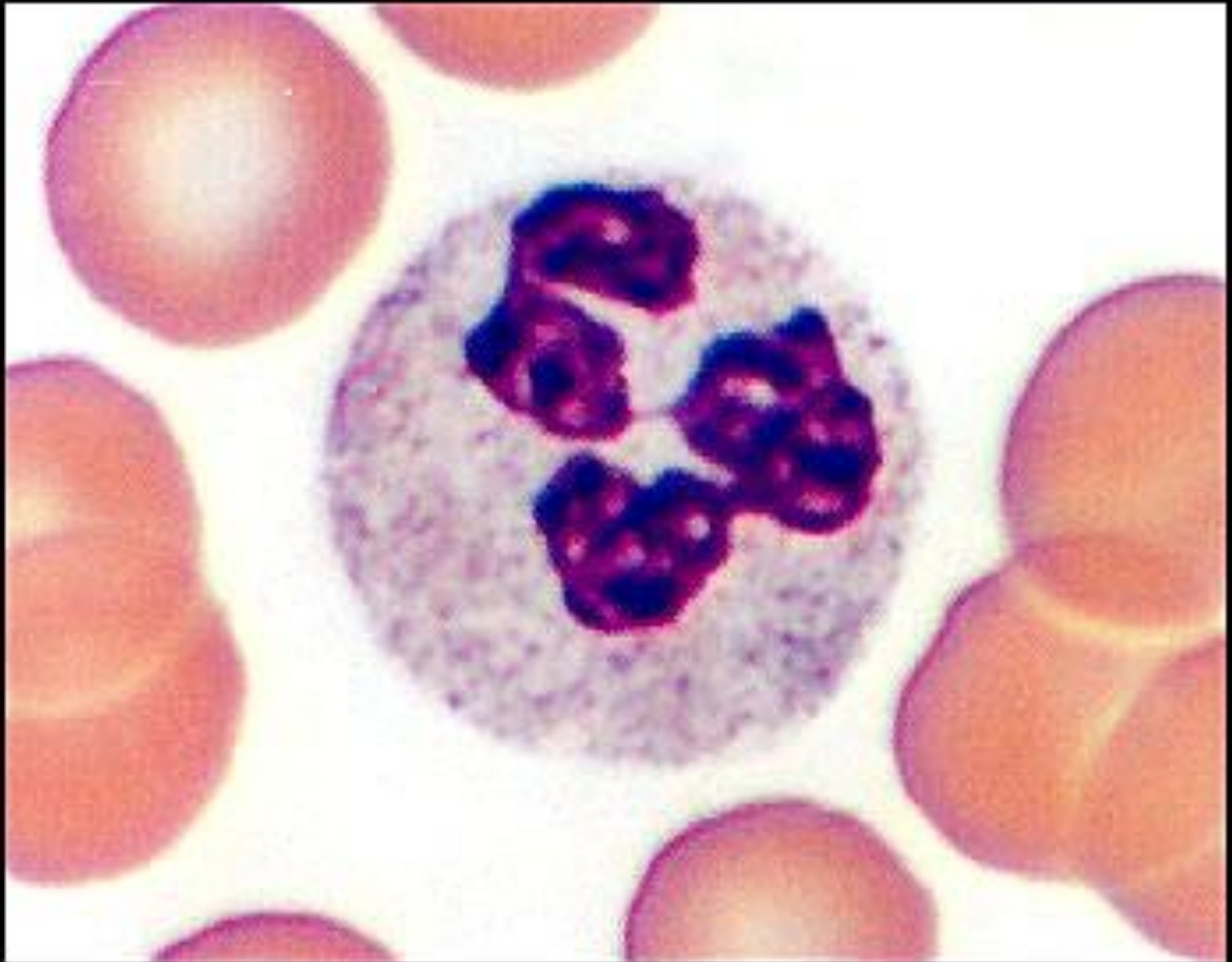
- All WBCs (leukocytes) have a nucleus and no hemoglobin
- Granular or agranular classification based on presence of cytoplasmic granules made visible by staining
 - granulocytes are neutrophils, eosinophils or basophils
 - agranulocytes are monocytes or lymphocytes

Leukocyte Production (Leukopoiesis)

- Committed cell types -- B & T progenitors and granulocyte-macrophage colony-forming units
 - possess receptors for colony-stimulating factors
 - released by mature WBCs in response to infections
- Red bone marrow stores & releases granulocytes & monocytes
- Some lymphocytes leave bone marrow unfinished
 - go to thymus to complete their development (T cells)
- Circulating WBCs do not stay in bloodstream
 - granulocytes leave in 8 hours & live 5 days longer
 - monocytes leave in 20 hours, transform into macrophages and live for several years
 - WBCs providing long-term immunity last decades

Neutrophils (Granulocyte)

- Polymorphonuclear Leukocytes or Polys
- Nuclei = 2 to 5 lobes connected by thin strands
 - older cells have more lobes
 - young cells called band cells because of horseshoe shaped nucleus (band)
- Fine, pale lilac practically invisible granules
- Diameter is 10-12 microns
- 60 to 70% of circulating WBCs



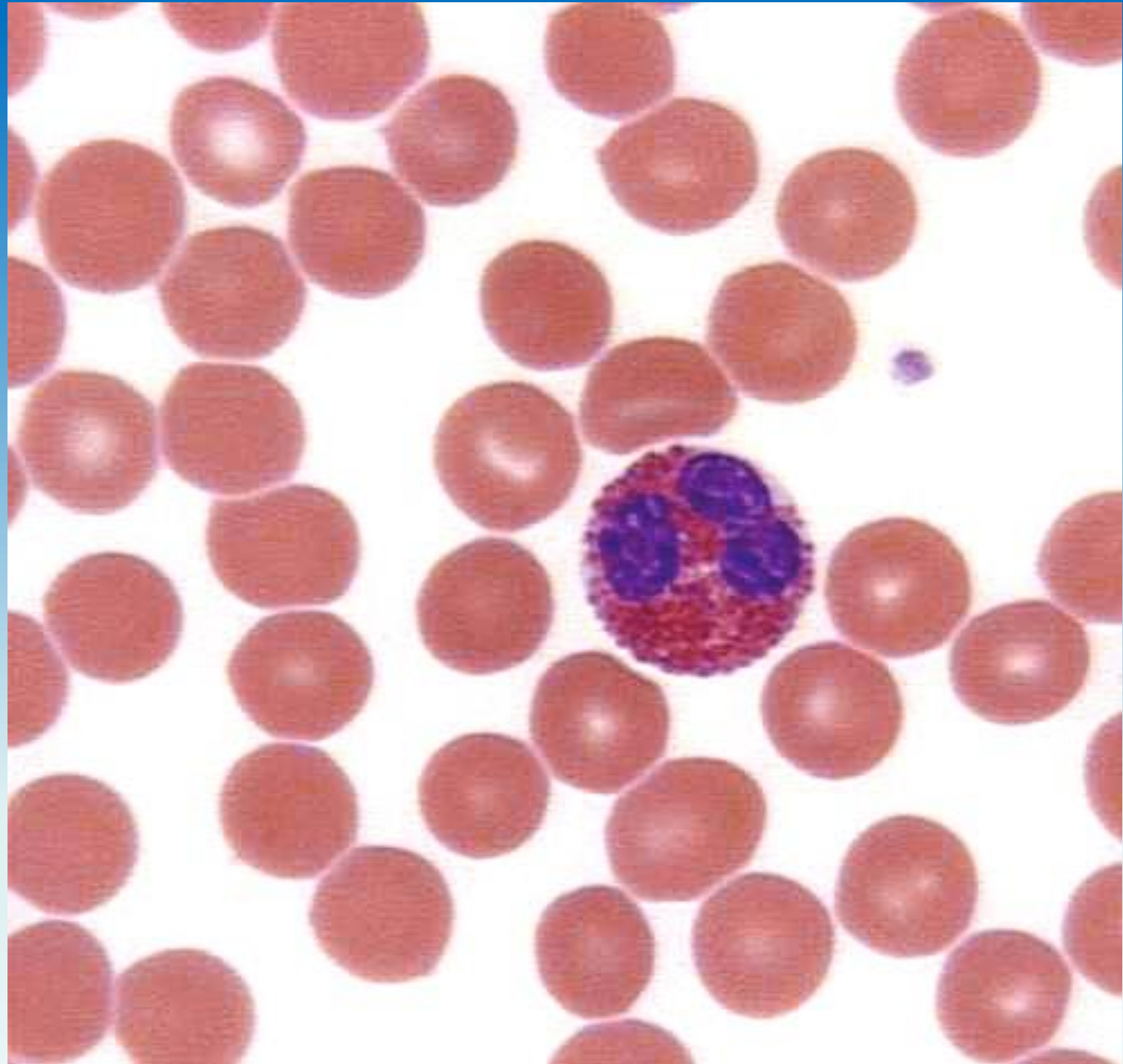


Neutrophil Function

- 60-70%
- Fastest response of all WBC to bacteria
- Direct actions against bacteria
 - release lysozymes which destroy/digest bacteria
 - release defensin proteins that act like antibiotics & poke holes in bacterial cell walls destroying them
 - release strong oxidants (bleach-like, strong chemicals) that destroy bacteria
- ↑ in bacterial infections

Eosinophils (Granulocyte)

- Nucleus with 2 or 3 lobes connected by a thin strand
- Large, uniform-sized granules stain orange-red with acidic dyes
 - do not obscure the nucleus
- Diameter is 10 to 12 microns
- 2 to 4% of circulating WBCs

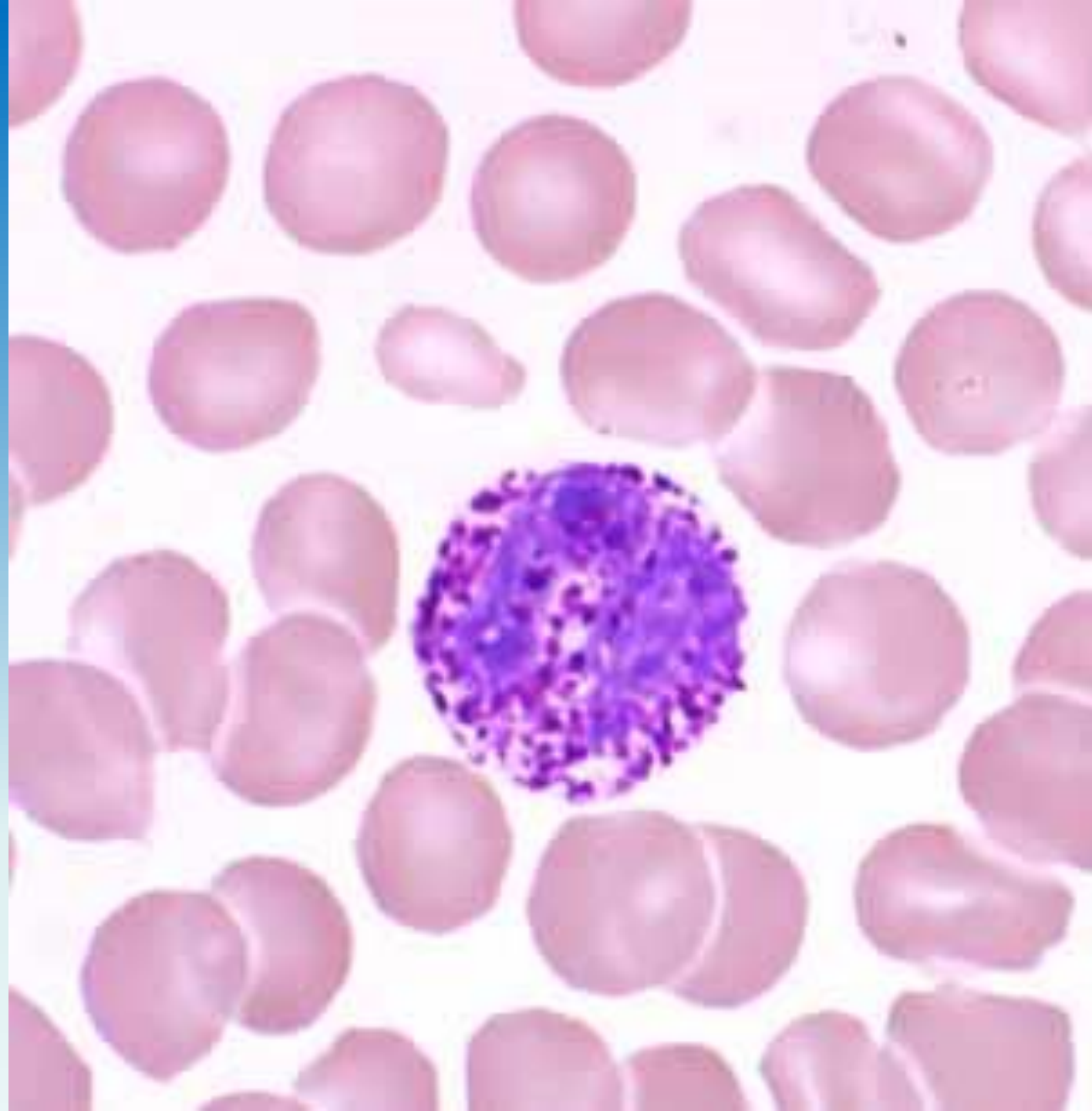


Eosinophil Function

- 2-4%
- Leave capillaries to enter tissue fluid
- Release histaminase
 - slows down inflammation caused by basophils
- Attack parasitic worms
- Phagocytize antibody-antigen complexes
- ↑ in parasitic infection and allergies

Basophils (Granulocyte)

- Large, dark purple, variable-sized granules stain with basic dyes
 - obscure the nucleus
- Irregular, s-shaped, bilobed nuclei
- Diameter is 8 to 10 microns
- Less than 1% of circulating WBCs

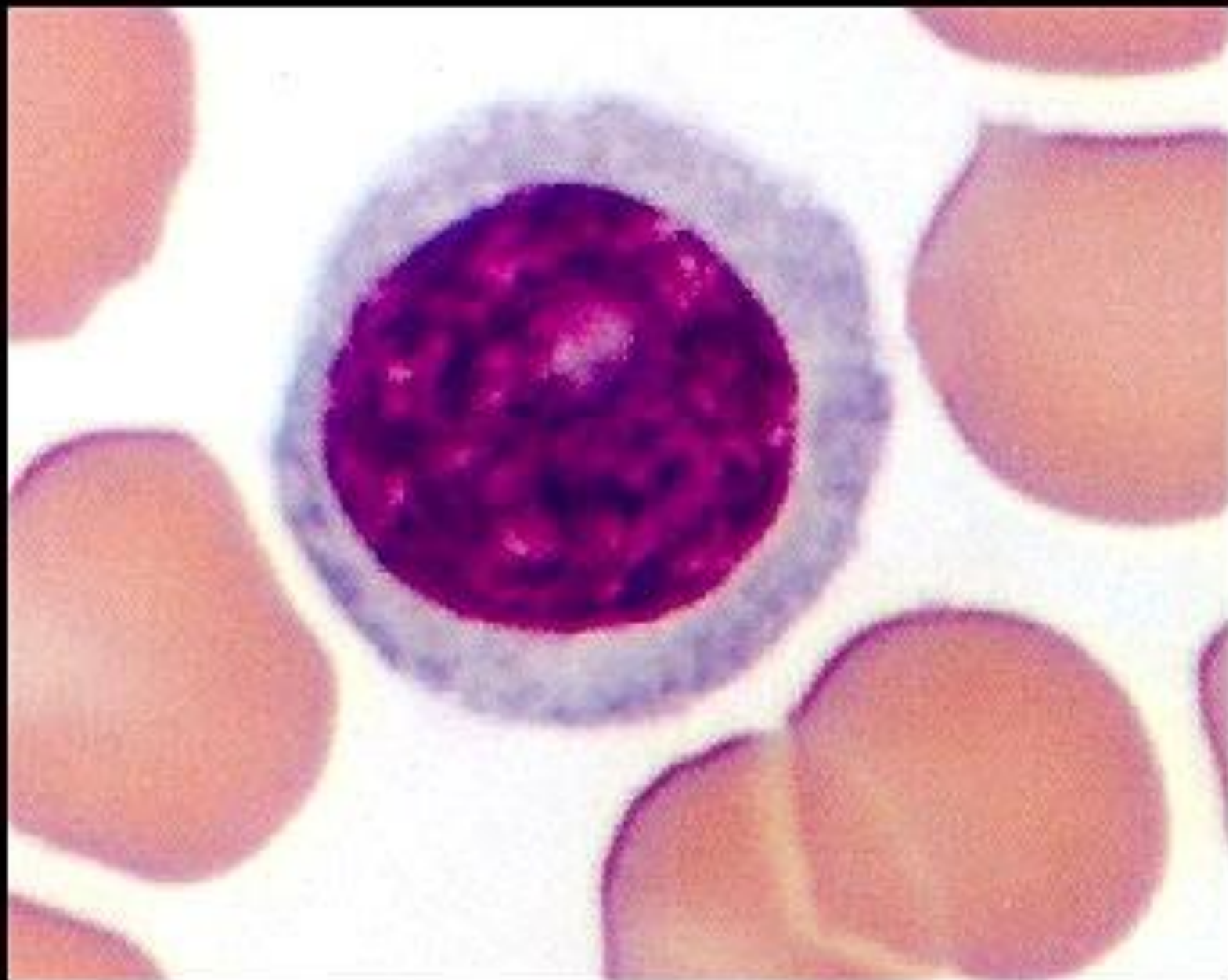


Basophil Function

- $<1\%$
- Involved in inflammatory and allergy reactions
- Leave capillaries & enter connective tissue as mast cells
- Release heparin, histamine & serotonin
 - heighten the inflammatory response and account for hypersensitivity (allergic) reaction
- $\uparrow$ in chicken pox, diabetes, autoimmune disorders, hypersensitivities (allergies)

Lymphocyte (Agranulocyte)

- Dark, oval to round nucleus
- Cytoplasm sky blue in color
 - amount varies from rim of blue to normal amount
- Small cells 6 - 9 microns in diameter
- Large cells 10 - 14 microns in diameter
 - increase in number during viral infections
- 20 to 25% of circulating WBCs



Lymphocyte Functions

- B cells
 - destroy bacteria and their toxins
 - turn into plasma cells that produces antibodies
 - Provide immunological memory
- T cells
 - attack viruses, fungi, transplanted organs, cancer cells & some bacteria
 - Present antigens to other immune cells
- Natural killer cells
 - attack many different microbes & some tumor cells
 - destroy foreign invaders by direct attack
- ↑ in diverse infections and immune responses

Monocyte (Agranulocyte)

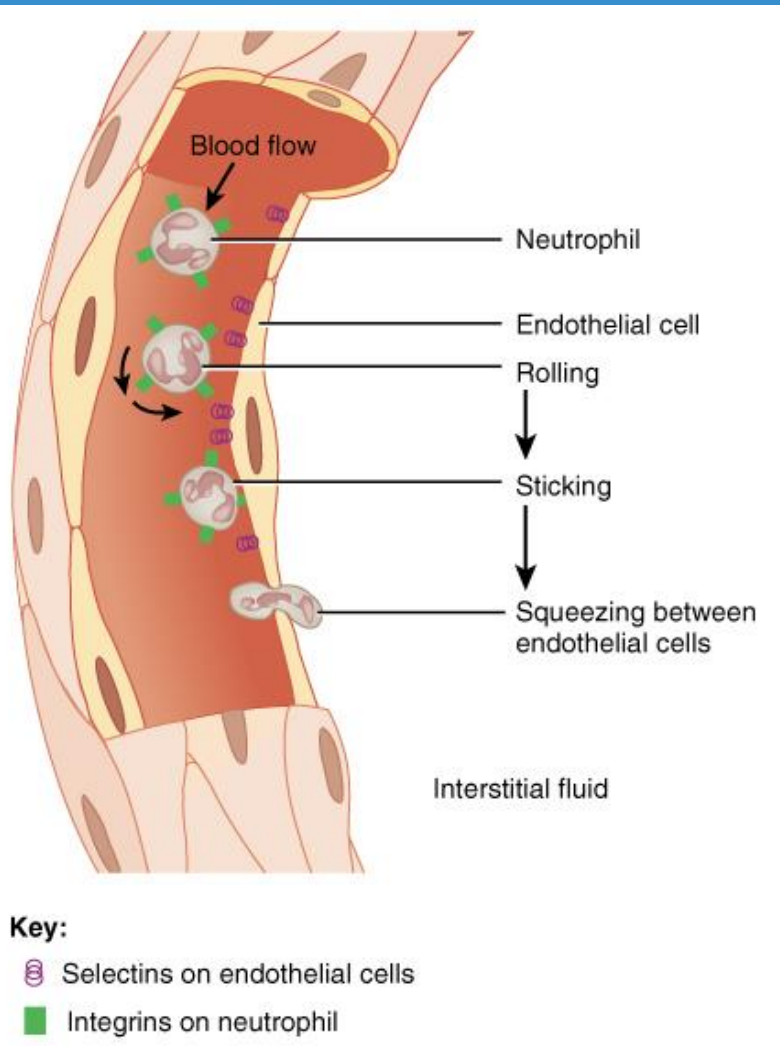
- Nucleus is kidney or horse-shoe shaped
- Largest WBC in circulating blood
 - does not remain in blood long before migrating to the tissues
 - differentiate into macrophages
 - fixed group found in specific tissues
 - alveolar macrophages in lungs
 - kupffer cells in liver
 - wandering group gathers at sites of infection
- Diameter is 12 - 20 microns
- Cytoplasm is a foamy blue-gray
- 3 to 8% of circulating WBCs



Monocyte Function

- 8%
- Take longer to get to site of infection, but arrive in larger numbers
- Become wandering macrophages, once they leave the capillaries
- Destroy microbes and clean up dead tissue following an infection
- ↑ in viral infections and inflammation

Emigration & Phagocytosis in WBCs



- WBCs roll along endothelium, stick to it & squeeze between cells.
 - adhesion molecules (selectins) help WBCs stick to endothelium
 - displayed near site of injury
 - molecules (integrins) found on neutrophils assist in movement through wall
- Neutrophils & macrophages phagocytize bacteria & debris
 - chemotaxis of both
 - kinins from injury site & toxins

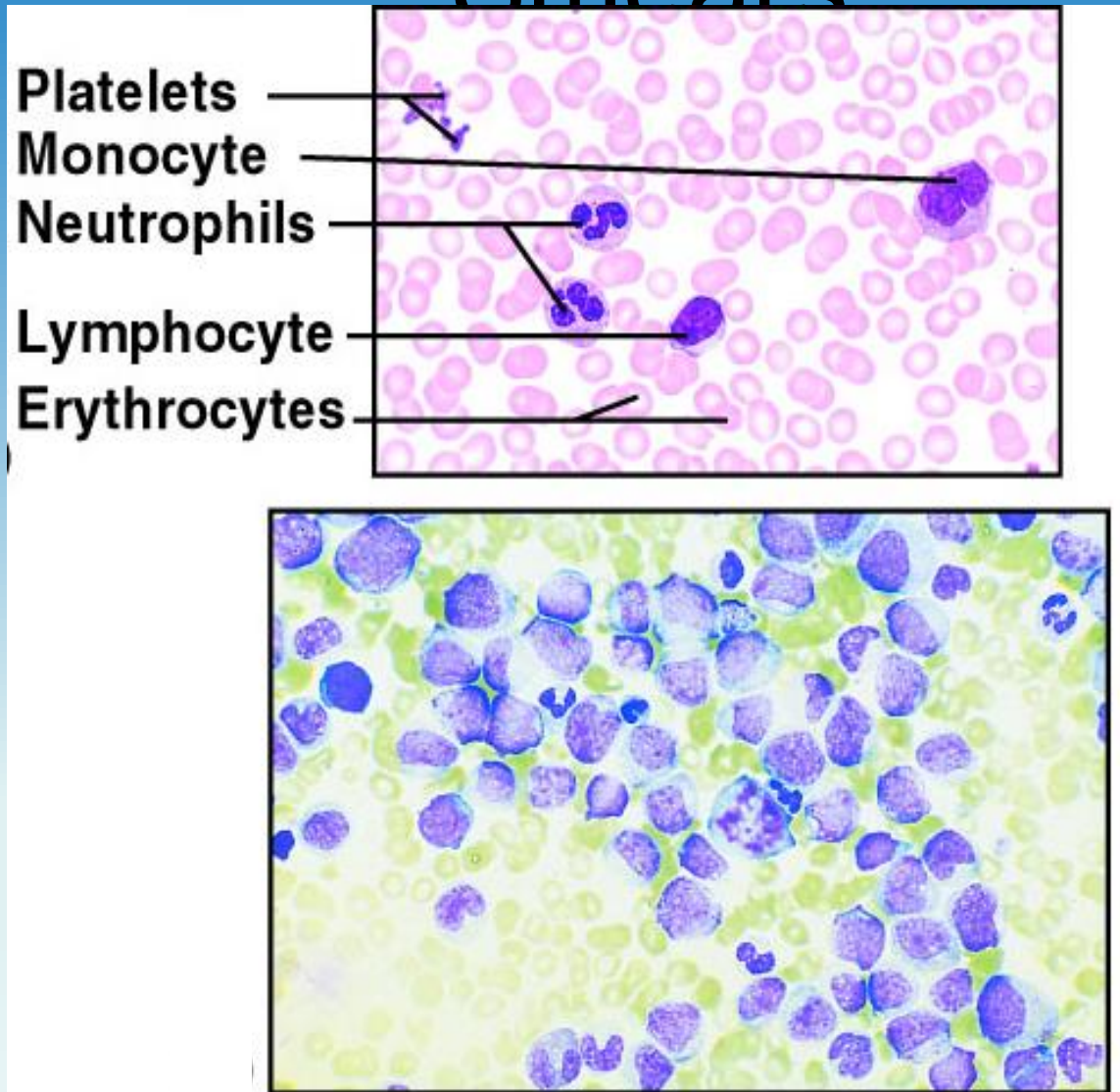
Differential WBC Count

- Detection of changes in numbers of circulating WBCs (percentages of each type)
 - indicates infection, poisoning, leukemia, chemotherapy, parasites or allergy reaction
- Normal WBC counts
 - neutrophils 60-70% (up if bacterial infection)
 - lymphocyte 20-25% (up if viral infection)
 - monocytes 3 -- 8 % (up if fungal/viral infection)
 - eosinophil 2 -- 4 % (up if parasite or allergy reaction)
 - basophil <1% (up if allergy reaction or hypothyroid)

Abnormalities of Leukocyte Count

- Leukopenia = low WBC count ($<5000/\mu\text{L}$)
 - causes -- radiation, poisons, infectious disease
 - effects -- elevated risk of infection
- Leukocytosis = high WBC count ($>10,000/\mu\text{L}$)
 - causes -- infection, allergy & disease
 - differential count -- distinguishes % of each cell type
- Leukemia = cancer of hemopoietic tissue
 - myeloid and lymphoid -- uncontrolled WBC production
 - acute and chronic -- death in either months or $\cong 3$ years
 - effects -- normal cell % disrupted, patient subject to opportunistic infection, anemia & impaired clotting

Normal and Leukemia Blood Smears



Leukemia

- Acute leukemia
 - uncontrolled production of immature leukocytes
 - crowding out of normal red bone marrow cells by production of immature WBC
 - prevents production of RBC & platelets
- Chronic leukemia
 - accumulation of mature WBC in bloodstream because they do not die
 - classified by type of WBC that is predominant---monocytic, lymphocytic.

Bone Marrow Transplant

- Intravenous transfer of healthy bone marrow
- Procedure
 - destroy sick bone marrow with radiation & chemotherapy
 - donor matches surface antigens on WBC
 - put sample of donor marrow into patient's vein for reseeded of bone marrow
 - success depends on histocompatibility of donor & recipient
- Treatment for leukemia, sickle-cell, breast, ovarian or testicular cancer, lymphoma or aplastic anemia

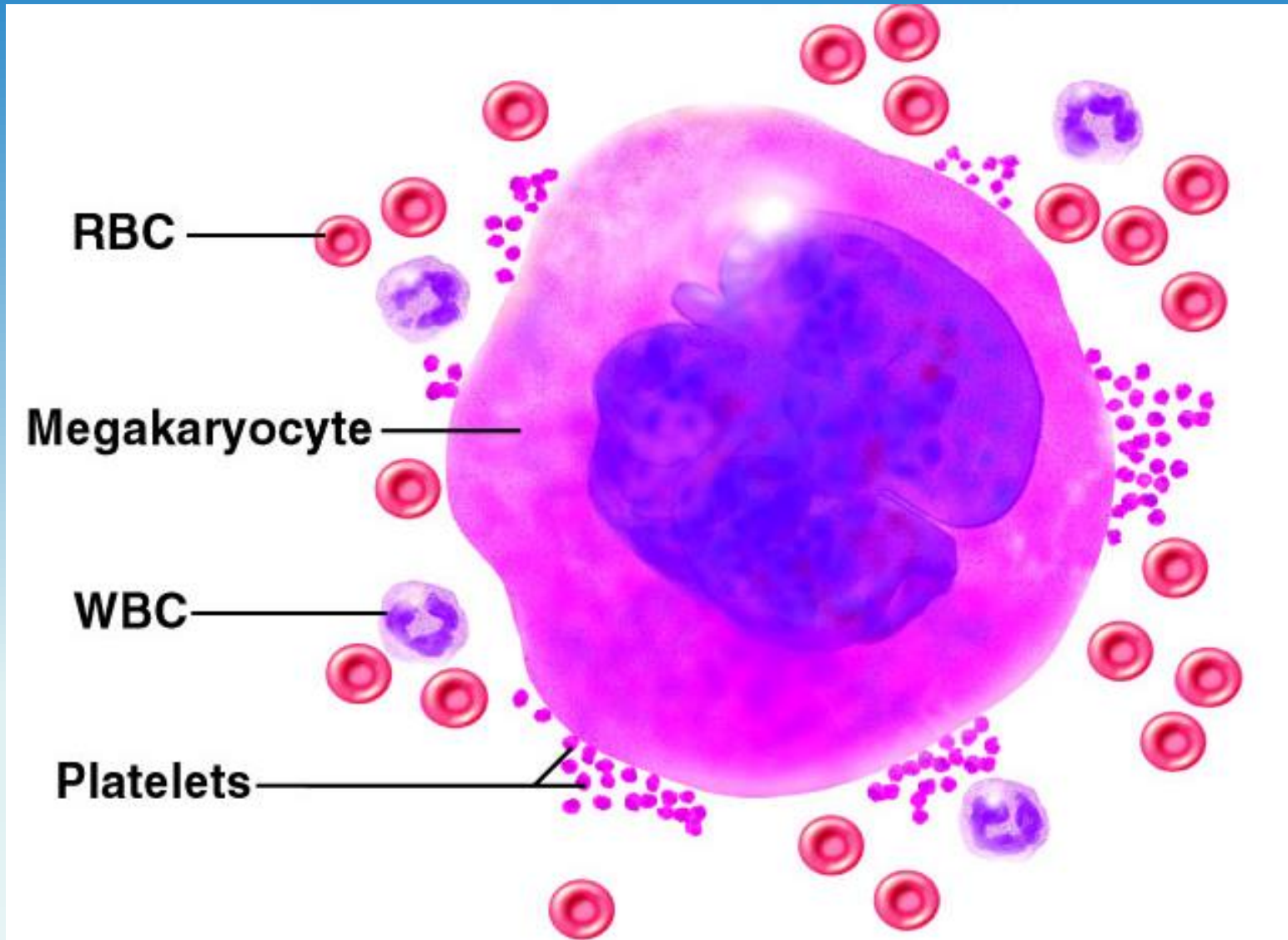
Platelet (Thrombocyte) Anatomy

- Disc-shaped, 2 - 4 micron cell fragment with no nucleus
- Normal platelet count is 150,000-400,000/drop of blood
- Other blood cell counts
 - 5 million red & 5-10,000 white blood cells

Platelet Production (Thrombopoiesis)

- Hemocytoblast that develops receptors for thrombopoietin from liver or kidney becomes megakaryoblast
- Megakaryoblast repeatedly replicates its DNA without dividing
 - forms gigantic cell called megakaryocyte (100 μm in diameter that remains in bone marrow)
- Infoldings of megakaryocyte cytoplasm splits off cell fragments that enter the bloodstream as platelets (live for 10 days)
 - some stored in spleen & released as needed

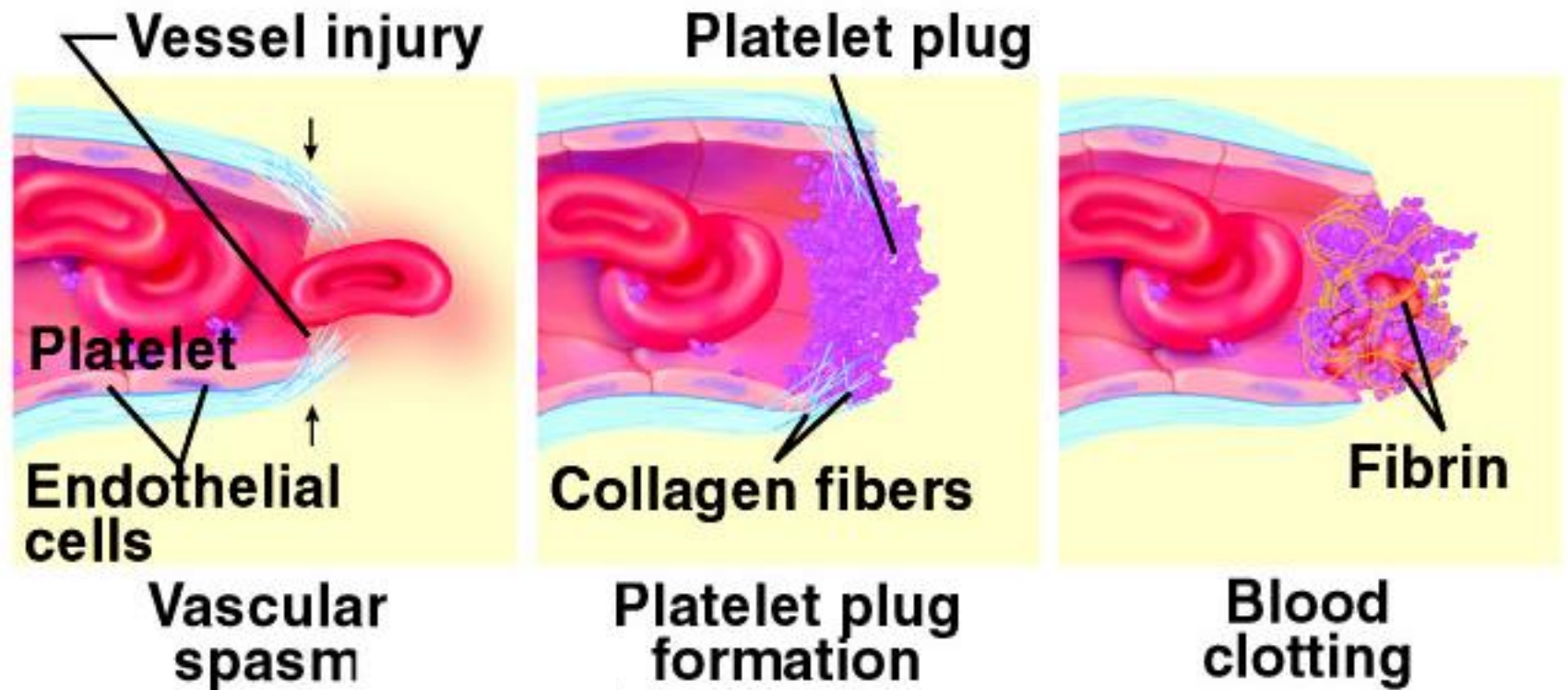
Megakaryocytes & Platelets



Platelets

- Small fragments of megakaryocyte cytoplasm
 - 2-4 μm diameter & containing “granules”
 - pseudopods provide amoeboid movement & phagocytosis
- Normal Count -- 130,000 to 400,000 platelets/ μL
- Functions
 - secrete clotting factors, growth factors for endothelial repair, and vasoconstrictors in broken vessels
 - form temporary platelet plugs
 - dissolve old blood clots
 - phagocytize bacteria
 - attract WBCs to sites of inflammation

Hemostasis



- Stopping bleeding in a quick & localized manner when blood vessels are damaged
- 3 hemostatic mechanisms all involve platelets

Vascular Spasm

- Prompt constriction of a broken vessel
- Triggers for a vascular spasm
 - some pain receptors directly innervate constrictors
 - lasts only a few minutes
 - injury to smooth muscle
 - longer-lasting constriction
 - platelets release serotonin, chemical vasoconstrictor
- Provides time for other 2 mechanisms to work

Platelet Plug Formation

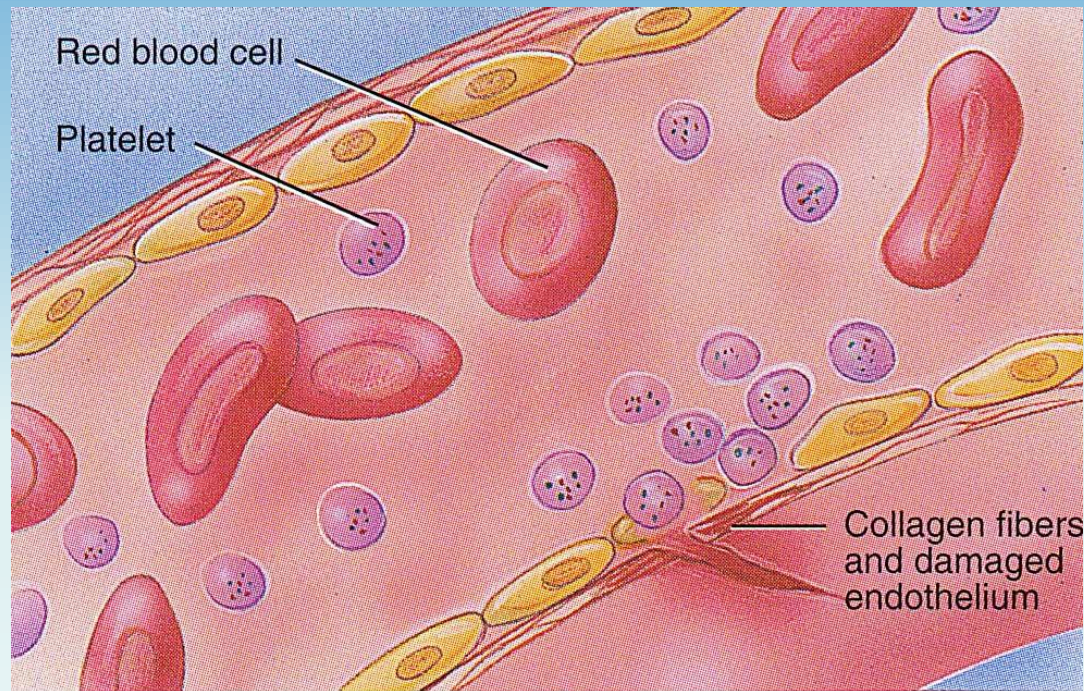
- Platelets store a lot of chemicals in granules needed for platelet plug formation
 - alpha granules
 - clotting factors
 - platelet-derived growth factor
 - cause proliferation of vascular endothelial cells, smooth muscle & fibroblasts to repair damaged vessels
 - dense granules
 - ADP, ATP, Ca^{+2} , serotonin, fibrin-stabilizing factor, & enzymes that produce thromboxane A_2
- Steps in the process
 - (1) platelet adhesion (2) platelet release reaction (3) platelet aggregation

Platelet Plug Formation

- Normal endothelium very smooth & coated with prostacyclin (platelet repellent)
- Broken vessel exposes rough surfaces of collagen
- Platelet plug formation begins
 - platelet pseudopods stick to damaged vessel and other platelets -- pseudopods contract and draw walls of vessel together forming a platelet plug
 - platelets degranulate releasing a variety of substances
 - serotonin is a vasoconstrictor
 - adenosine diphosphate (ADP) attracts & degranulates more platelets
 - thromboxane A₂, an eicosanoid that promotes aggregation, degranulation & vasoconstriction
- Positive feedback cycle is active until break in vessel is sealed

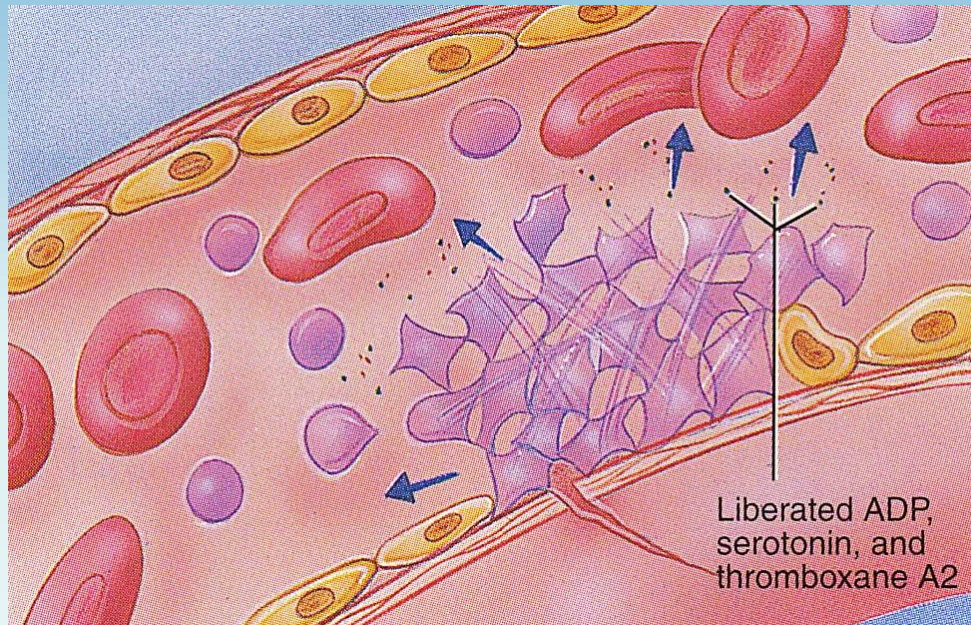
Platelet Adhesion

- Platelets stick to exposed collagen underlying damaged endothelial cells in vessel wall



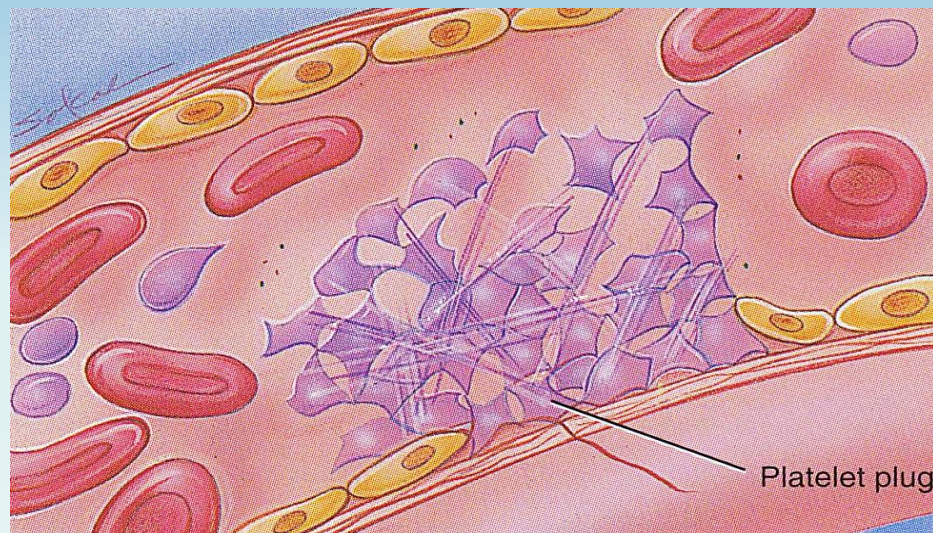
Platelet Release Reaction

- Platelets activated by adhesion
- Extend projections to make contact with each other
- Release thromboxane A₂ & ADP activating other platelets
- Serotonin & thromboxane A₂ are vasoconstrictors decreasing blood flow through the injured vessel



Platelet Aggregation

- Activated platelets stick together and activate new platelets to form a mass called a platelet plug
- Plug reinforced by fibrin threads formed during clotting process

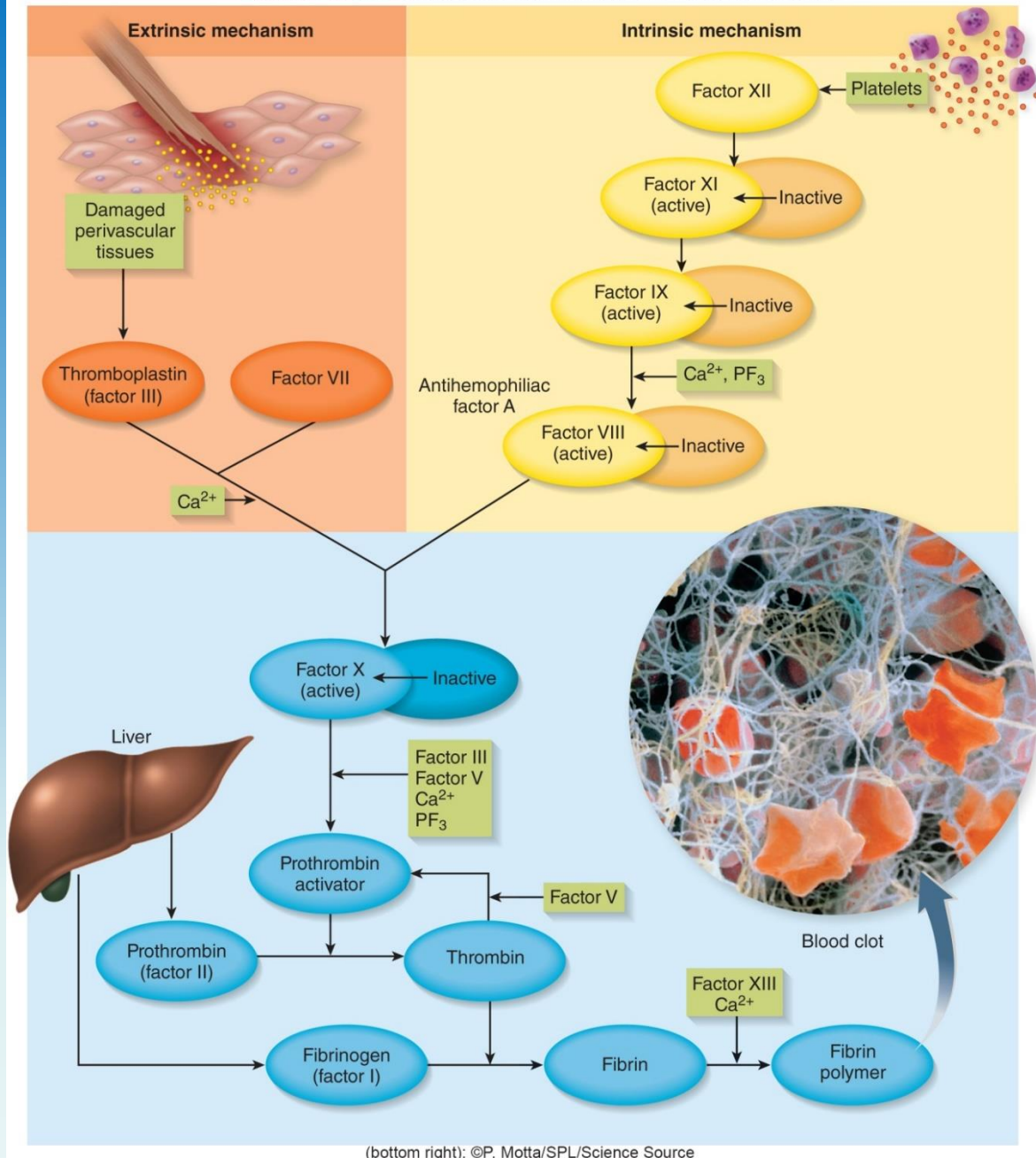


Coagulation

- Blood drawn from the body thickens into a gel
 - gel separates into liquid (serum) and a clot of insoluble fibers (fibrin) in which the cells are trapped
- If clotting occurs in an unbroken vessel is called a thrombosis
- Substances required for clotting are Ca^{+2} , enzymes synthesized by liver cells and substances released by platelets or damaged tissues
- Clotting is a cascade of reactions in which each clotting factor activates the next in a fixed sequence resulting in the formation of fibrin threads
 - prothrombinase & Ca^{+2} convert prothrombin into thrombin
 - thrombin converts fibrinogen into fibrin threads

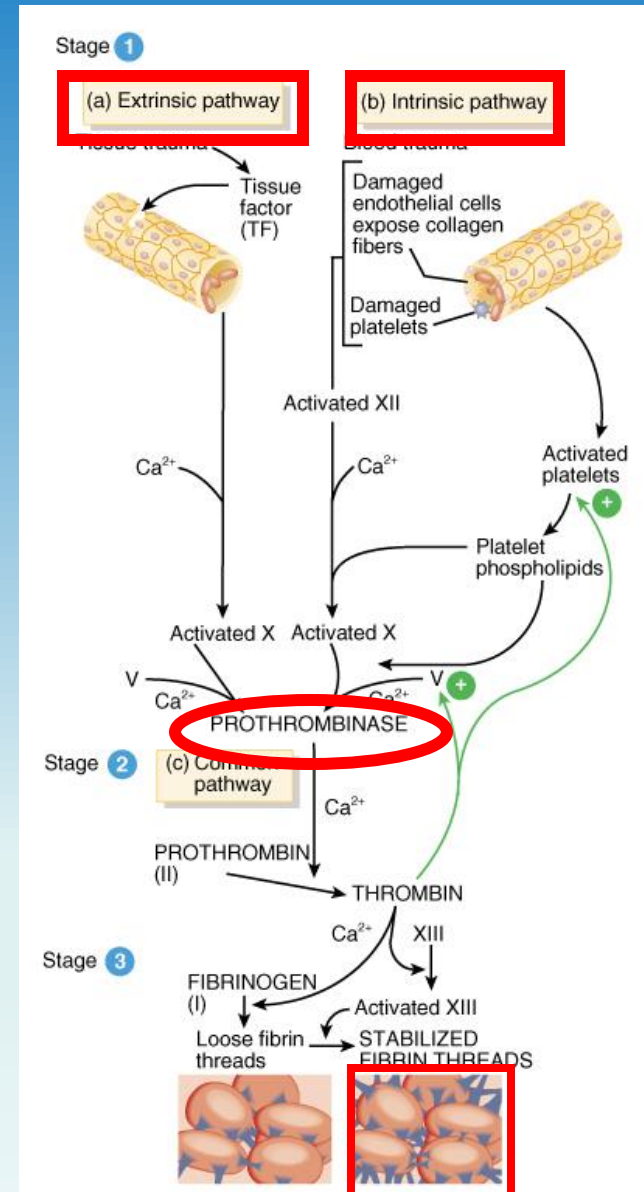
Coagulation

- Clotting is the most effective defense against bleeding --- needs to be quick but accurate
 - conversion of plasma protein fibrinogen into insoluble fibrin threads which form framework of clot
- Procoagulants or clotting factors (inactive form produced by the liver) are present in the plasma
 - activate one factor and it will activate the next to form a reaction cascade
- Factors released by the tissues cause the extrinsic cascade pathway to begin (damaged vessels)
- Factors found only in the blood itself causes the intrinsic cascade pathway to begin (platelet degranulation)
- Both cascades normally occur together



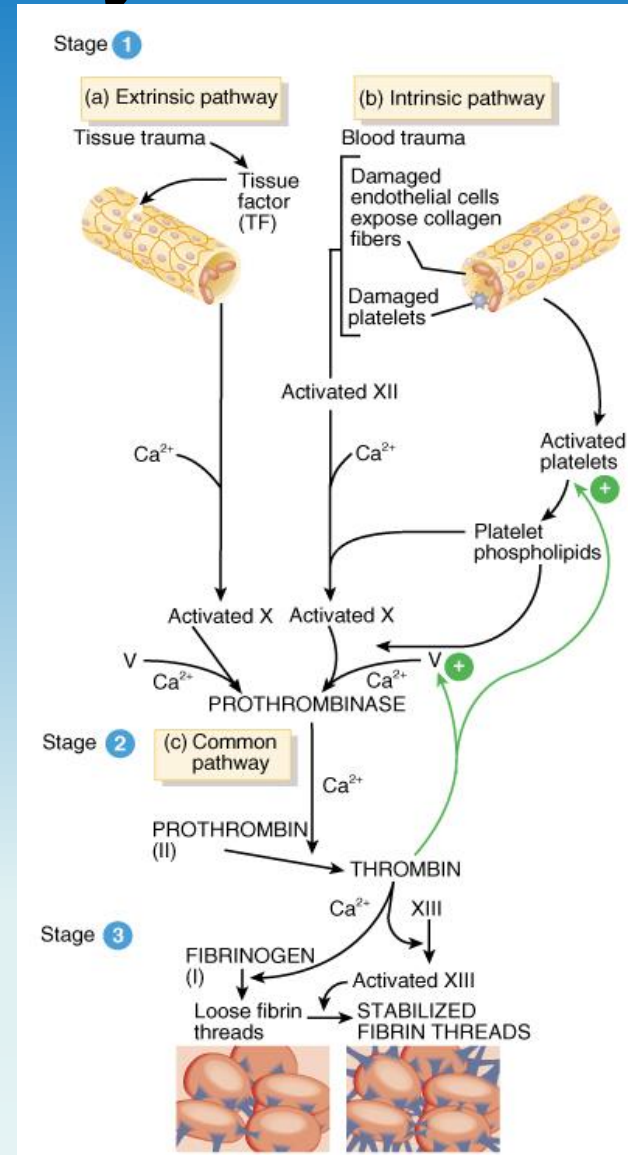
Overview of the Clotting Cascade

- Prothrombinase is formed by either the intrinsic or extrinsic pathway
- Final common pathway produces fibrin threads



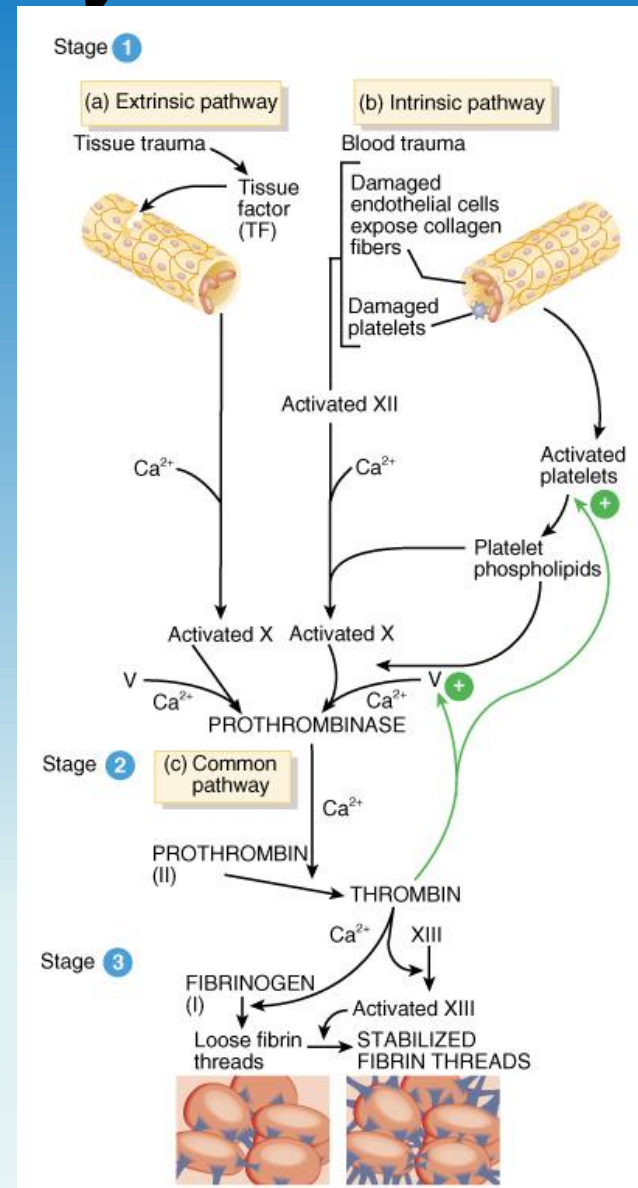
Extrinsic Pathway

- Damaged tissues leak tissue factor (thromboplastin Factor III to activate factor VII) into bloodstream
- Prothrombinase forms in seconds
- In the presence of Ca^{2+} , clotting factor X combines with V to form prothrombinase



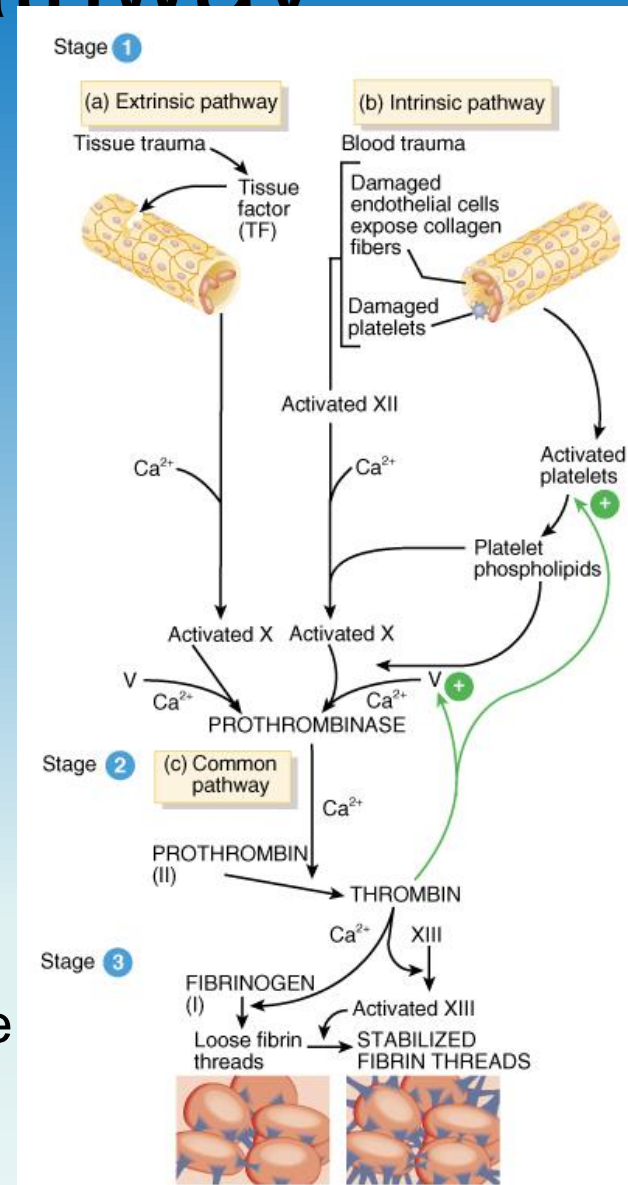
Intrinsic Pathway

- Activation occurs
 - endothelium is damaged & platelets come in contact with collagen of blood vessel wall activating Factor XII
 - platelets damaged & release phospholipids
- Requires several minutes for reaction to occur
- Substances involved: Ca^{+2} and clotting factors XII, X and V

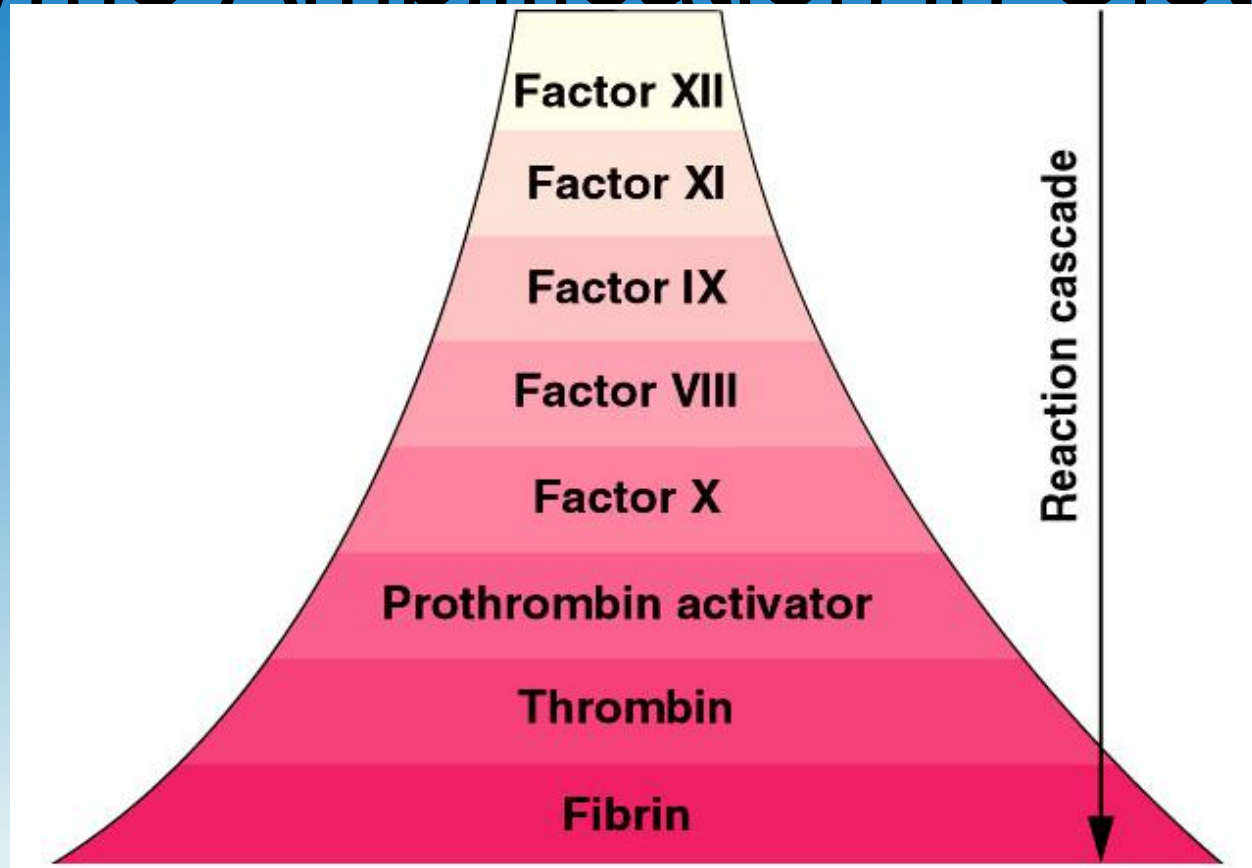


Final Common Pathway

- Prothrombinase and Ca^{2+}
 - catalyze the conversion of prothrombin to thrombin
- Thrombin
 - in the presence of Ca^{2+} converts soluble fibrinogen to insoluble fibrin threads
 - activates fibrin stabilizing factor XIII
 - positive feedback effects of thrombin
 - accelerates formation of prothrombinase
 - activates platelets to release phospholipids



Enzyme Amplification in Clotting



- Rapid clotting occurs since each activated enzyme produces a large number of enzyme molecules in the following step.

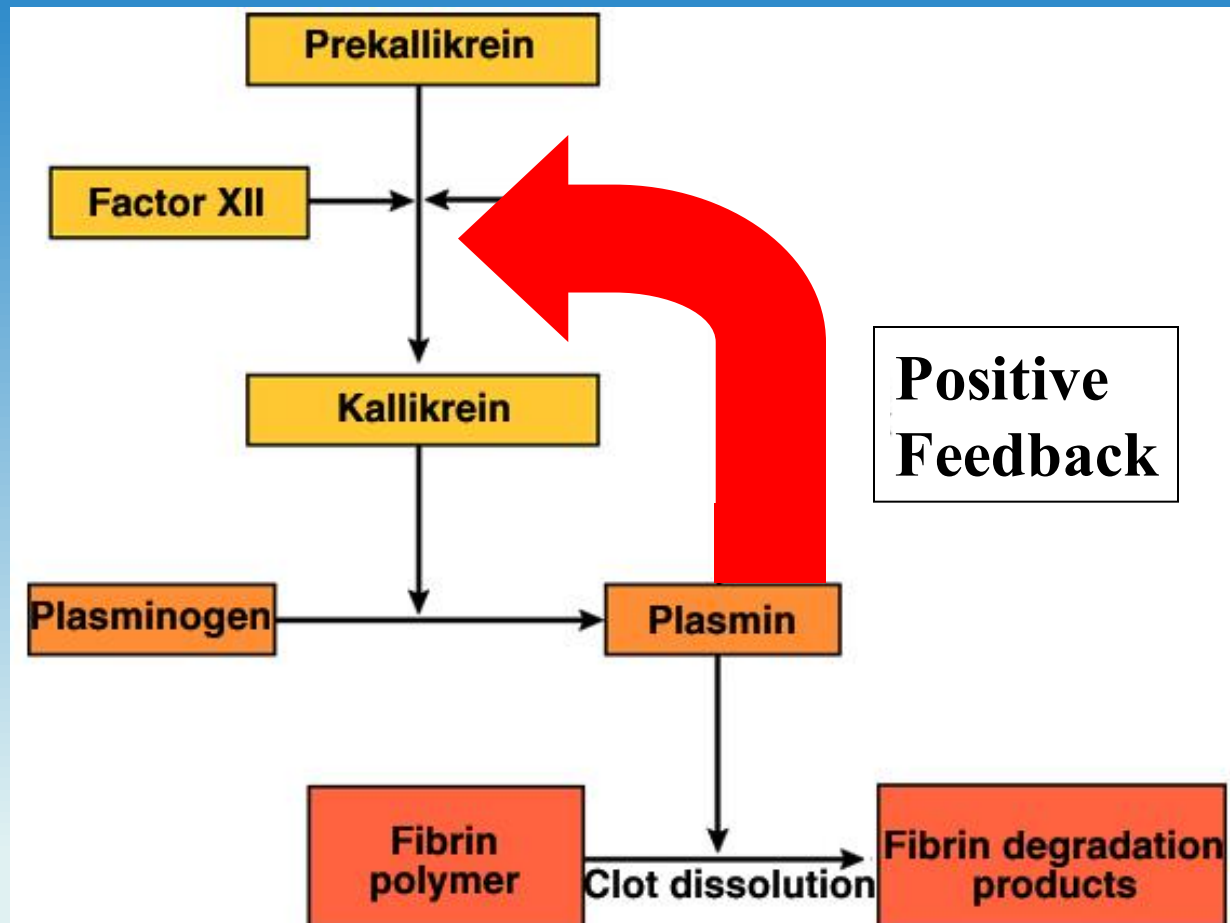
Role of Vitamin K in Clotting

- Normal clotting requires adequate vitamin K
 - fat soluble vitamin absorbed if lipids are present
 - absorption slowed if bile release is insufficient
- Required for synthesis of 4 clotting factors by hepatocytes
 - factors II (prothrombin), VII, IX and X
- Produced by bacteria in large intestine

The Fate of Blood Clots

- Clot retraction occurs within 30 minutes
 - trapped platelets release factor XIII stabilizing the fibrin threads
 - pseudopods of platelets contract pulling on fibrin threads condensing the clot and pulling the edges together
- Platelet-derived growth factor is secreted by platelets & endothelial cells
 - mitotic stimulant for fibroblasts and smooth muscle to multiply & repair the damaged vessel
- Fibrinolysis or dissolution of a clot
 - factor XII speeds up the formation of kallikrein enzyme
 - kallikrein converts plasminogen into plasmin, a fibrin-dissolving enzyme or clot buster

Blood Clot Dissolution



- Positive feedback occurs
- Plasmin promotes formation of kallikrein

Hemostatic Control Mechanisms

- Fibrinolytic system dissolves small, inappropriate clots & clots at a site of a completed repair
 - fibrinolysis is dissolution of a clot
- Inactive plasminogen is incorporated into the clot
 - activation occurs because of factor XII and thrombin
 - plasminogen becomes plasmin (fibrinolysin) which digests fibrin threads
- Clot formation remains localized
 - fibrin absorbs thrombin
 - Rapid blood flow dilutes clotting factors
 - endothelial cells & WBC produce prostacyclin that opposes thromboxane A₂ (platelet adhesion & release)
- Natural anticoagulants
 - antithrombin produced by the liver deactivates thrombin before it can act on fibrinogen
 - heparin secreted by basophils & mast cells interferes with formation of prothrombin activator

Intravascular Clotting

- Thrombosis
 - clot (thrombus) forming in an unbroken blood vessel
 - forms on rough inner lining of BV
 - if blood flows too slowly (stasis) allowing clotting factors to build up locally & cause coagulation
 - may dissolve spontaneously or dislodge & travel
- Embolus
 - clot, air bubble or fat from broken bone in the blood
 - pulmonary embolus is found in lungs
- Low dose aspirin blocks synthesis of thromboxane A₂ & reduces inappropriate clot formation
 - strokes, TIAs and myocardial infarctions

Anticoagulants and Thrombolytic Agents

- Anticoagulants suppress or prevent blood clotting
 - heparin
 - administered during hemodialysis and surgery
 - warfarin (Coumadin)
 - antagonist to vitamin K so blocks synthesis of clotting factors
 - slower than heparin
 - stored blood in blood banks treated with citrate phosphate dextrose (CPD) that removes Ca^{+2}
- Thrombolytic agents are injected to dissolve clots
 - directly or indirectly activate plasminogen
 - streptokinase or tissue plasminogen activator (t-PA)

Medicinal Leeches Removing Clots

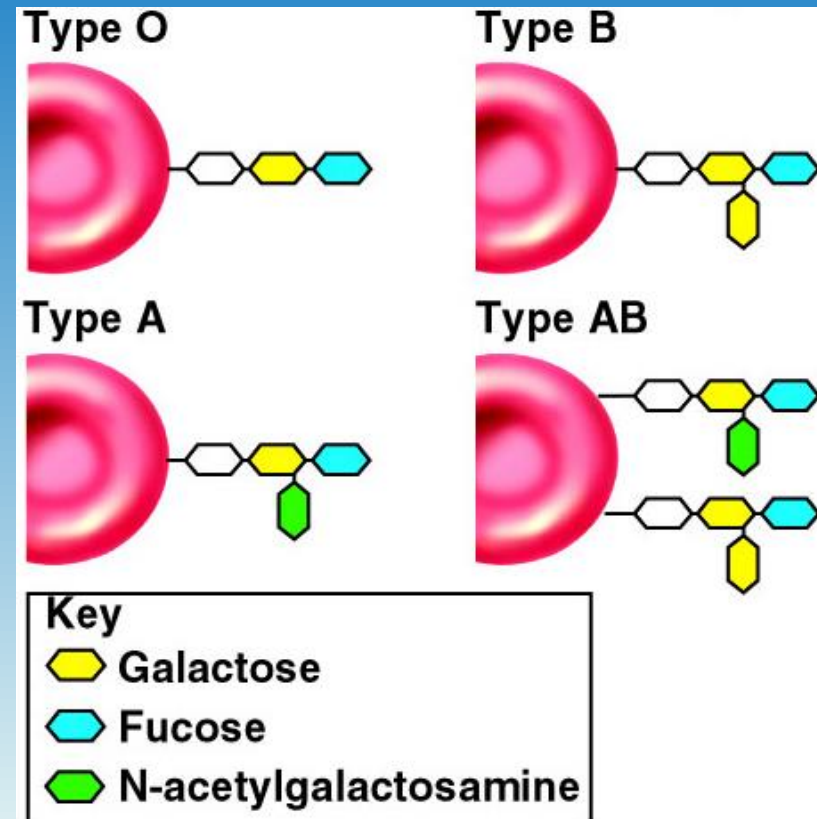


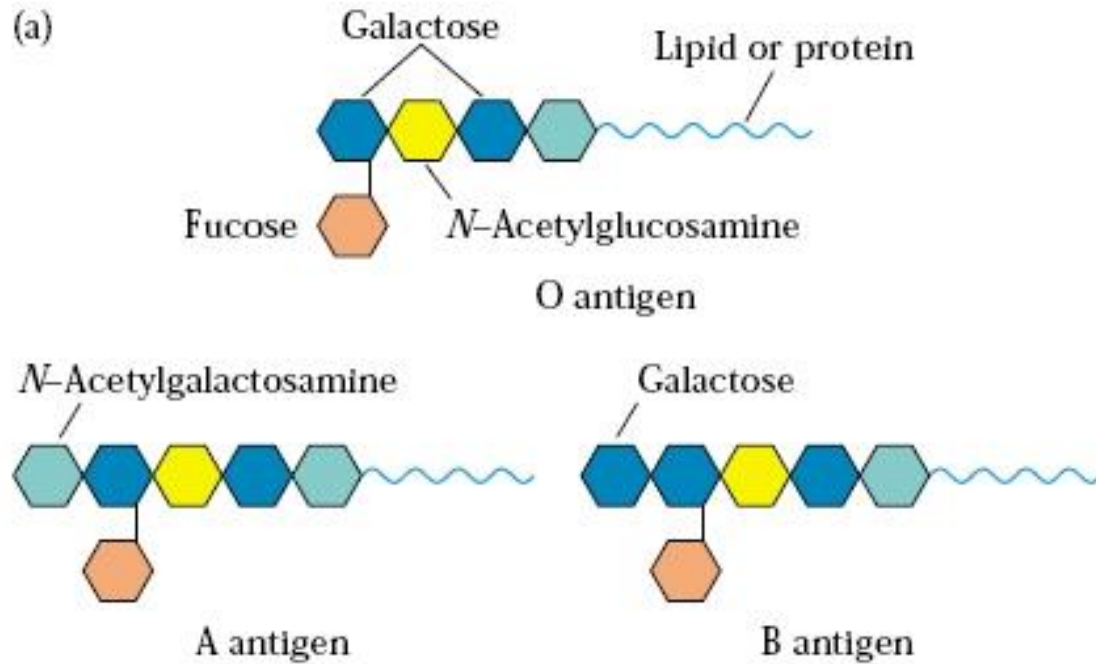
Hemophilia

- genetic lack of any clotting factor affects coagulation
- sex-linked recessive in males (inherit from mother)
 - hemophilia A is missing factor VIII (83% of cases,)
 - hemophilia B is missing factor IX (15% of cases)
- hemophilia C is missing factor XI (autosomal)
 - Less severe because alternate clotting activator exists
- physical exertion causes bleeding & excruciating pain
 - transfusion of plasma or purified clotting factors
 - factor VIII now produced by transgenic bacteria

Blood Types

- RBC antigens
 - called agglutinogens A & B
 - inherited combinations of proteins, glycoproteins and glycolipids on red blood cell
 - At least 26 different groups ABO, Rh, Kell, Lewis, Kidd, Duffy, etc...
- Plasma antibodies
 - called agglutinins anti-A & -B
 - gamma globulins in blood plasma that recognize (stick to) foreign agglutinogens on RBCs
 - responsible for RBC agglutination in mismatched blood transfusions





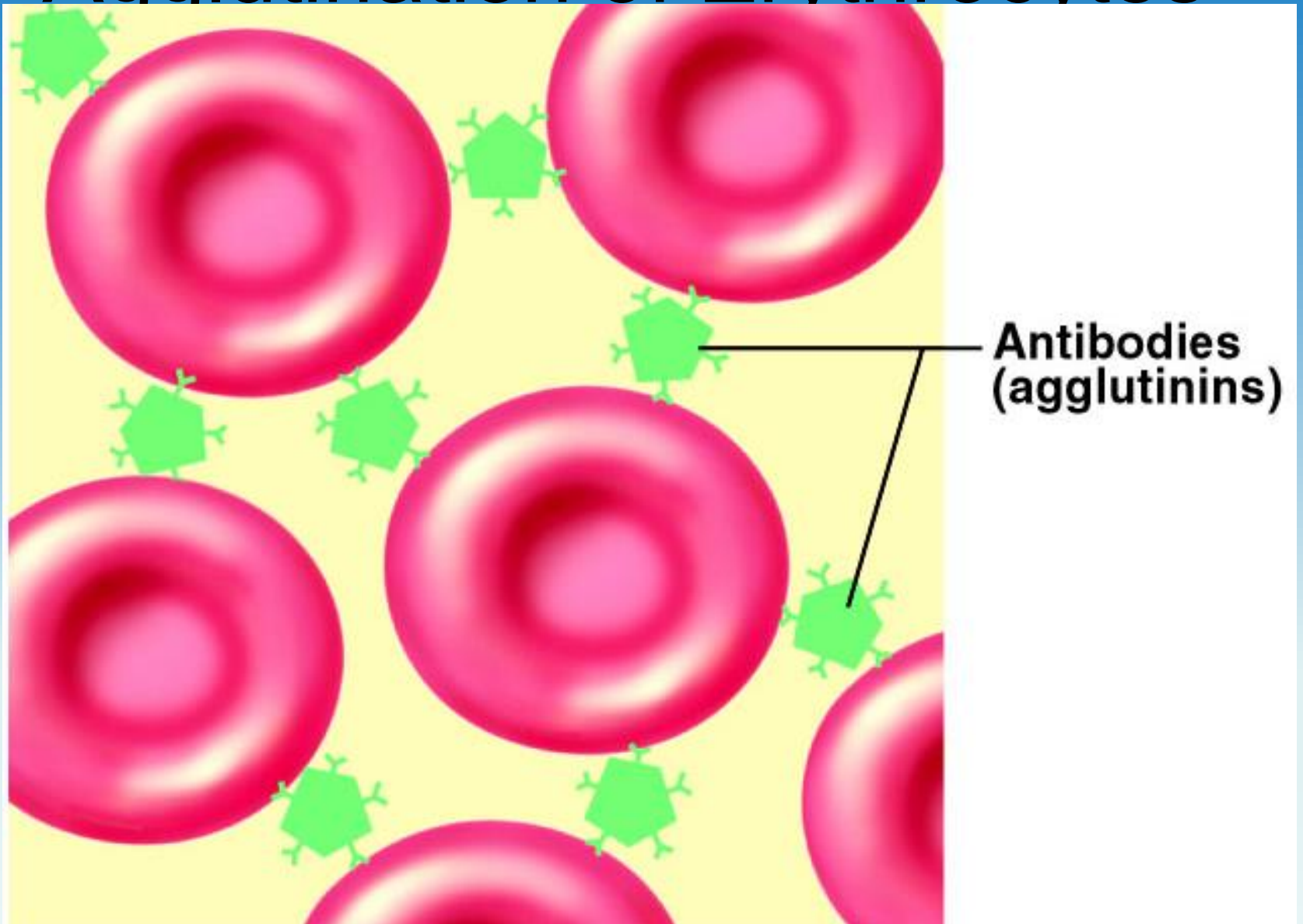
(b)

Genotype	Blood-group phenotype	Antigens on erythrocytes (<i>agglutinins</i>)	Serum antibodies (<i>isohemagglutinins</i>)
AA or AO	A	A	Anti-B
BB or BO	B	B	Anti-A
AB	AB	A and B	None
OO	O	None	Anti-A and anti-B

The ABO Group

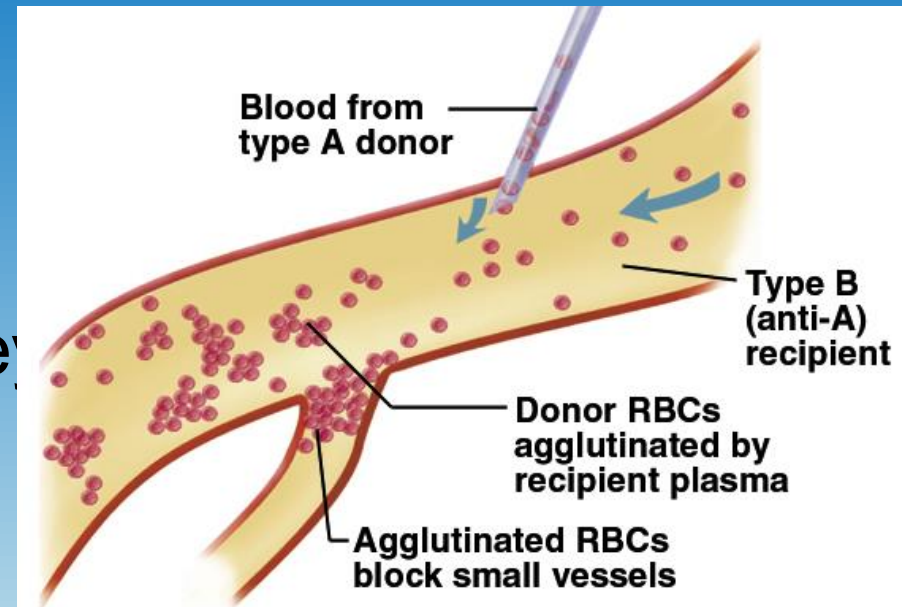
- Your ABO blood type is determined by presence or absence of antigens (agglutinogens) A & B on RBCs
 - blood type A person has A antigens, blood type B person has B antigens, AB has both & blood type O has neither
 - blood type O is the most common; AB the rarest
- Antibodies (agglutinins) appear 2-8 months after birth & are at maximum concentration at 10 yr.
 - antibodies A and/or B, both or none are in plasma
 - you do not have those that would react against your own antigens
 - each antibody can attach to several antigens at the same time causing agglutination (clumping)

Agglutination of Erythrocytes



Mismatched Transfusion Reaction

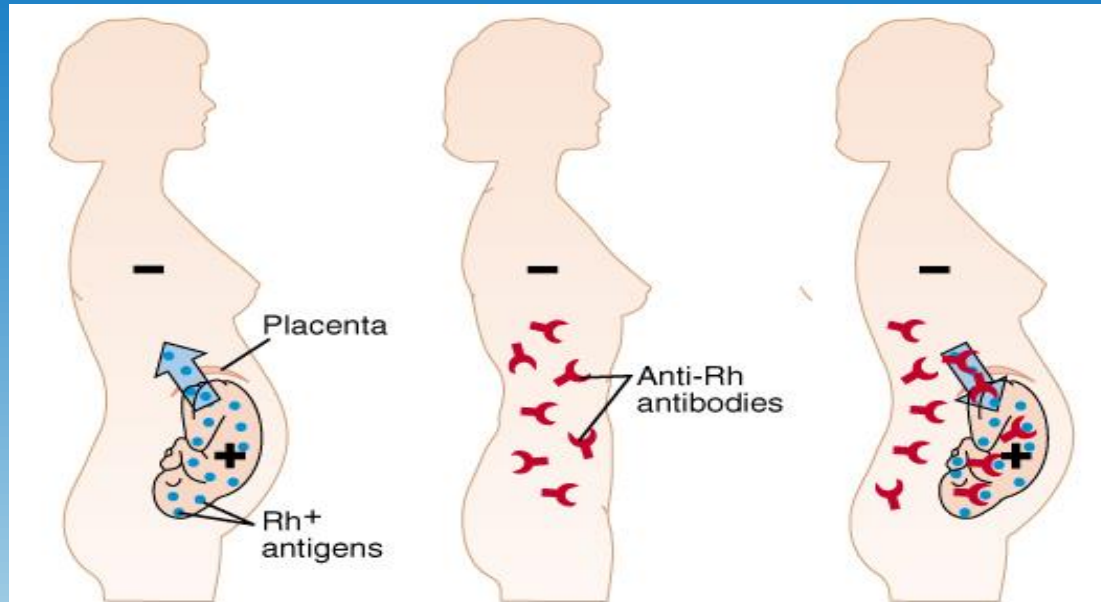
- Agglutinated RBCs block blood vessels & rupture
 - free Hb can block kidney tubules & cause death
- Universal donors and recipients
 - AB called universal recipient since it lacks both antibody A and B; O called universal donor
 - problem is donor's plasma may have antibodies against recipient's red blood cells
 - solution is giving packed cells with minimum plasma



The Rh Group

- Rh or D agglutinogens discovered in rhesus monkey in 1940
 - blood type is Rh+ if agglutinogens present on RBCs
 - Rh frequencies vary among ethnic groups
- Anti-D agglutinins are not normally present in blood
 - form only in individuals exposed to Rh+ blood
 - Rh- pregnant woman carrying an Rh+ fetus or blood transfusion of Rh+ blood
 - no problems result with either the **first** transfusion or the **first** pregnancy, abortion or miscarriage
 - hemolytic disease of the newborn (erythroblastosis fetalis) occurs if mother has formed antibodies & is pregnant with **2nd** Rh+ child or **2nd** transfusion
 - RhoGAM is given to pregnant woman to prevent antibody formation and prevent any future problems
 - RhoGAM binds fetal agglutinogens in her blood so she will not form antibodies against them during the pregnancy

Hemolytic Disease of Newborn



- Rh negative mom and Rh+ fetus will have mixing of blood at birth
- Mom's body creates Rh antibodies unless she receives a RhoGam shot soon after first delivery, miscarriage or abortion
 - RhoGam binds to loose fetal blood and removes it from body before she reacts
- In subsequent Rh+ pregnancies mother's antibodies attack fetal blood causing severe anemia & toxic brain syndrome from excessive bilirubin in blood
 - treatment is phototherapy to degrade bilirubin or exchange transfusion to completely replace infant's blood

Transfusion and Transfusion Reactions

- Transfer of whole blood, cells or plasma into the bloodstream of recipient
 - used to treat anemia or severe blood loss
- Incompatible blood transfusions
 - antigen-antibody complexes form between plasma antibodies & “foreign proteins” on donated RBC's (agglutination)
 - donated RBCs become leaky (complement proteins) & burst
 - loose hemoglobin causes kidney damage
- Problems caused by incompatibility between donor's cells and recipient's plasma
- Donor plasma is too diluted to cause problems

Universal Donors and Recipients

- People with type AB blood called “universal recipients” since have no antibodies in plasma
 - only true if cross match the blood for other antigens
- People with type O blood cell called “universal donors” since have no antigens on their cells
 - theoretically can be given to anyone

Disseminated Intravascular Clotting

- Life threatening paradoxical presence of blood clotting and bleeding at the same time throughout the whole body
 - so many clotting factors are removed by widespread clotting that too few remain to permit normal clotting
- Associated with infections, hypoxia, low blood flow rates, trauma, hypotension & hemolysis
- Clots cause ischemia and necrosis leading to multisystem organ failure